

Nonreal Roots Survive in Polynomial Iterative Equations

A cubic counterexample and a classification of circular spectra

Part II: the increasing spectral core on several circles

Part III: orientation-reversing spectra

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Abstract

For a real polynomial $P(z) = \sum_{j=0}^d a_j z^j$, write $P[f](x) = \sum_{j=0}^d a_j f^{\circ j}(x)$, where $f^{\circ 0}(x) = x$. A published question of Draga and Morawiec asks whether nonreal characteristic roots of maximal modulus can be removed when a real root has the same modulus. We give a negative answer under the stated continuity assumptions. An explicit, odd, increasing, globally bi-Lipschitz homeomorphism F satisfies $F^{\circ 3}(x) = 8x$, although $F(1) = 3$. Its minimal iterative polynomial is $z^3 - 8$, certified by a 3×3 Hankel determinant equal to -56 . A second construction is real analytic at every real point and has minimal polynomial $(z - 1)^2(z^2 + 1)$; deleting its nonreal roots also fails.

We then determine exactly which monic real polynomials supported on one circle $|z| = r > 0$ occur as minimal iterative polynomials of increasing homeomorphisms of $\mathbb{R}$. For $r \neq 1$, the positive root r must have odd multiplicity, at least as large as every other multiplicity. For $r = 1$, the identity is exceptional; otherwise the multiplicity of 1 must be even and strictly larger than every other multiplicity. Every permitted polynomial is realized by an explicit small trigonometric perturbation of a polynomial–exponential coordinate. The proof uses two-sided orbit recurrences and an elementary positivity lemma. We also obtain sharp increasing-solution rigidity criteria and explain the role of differentiability at a fixed point. Exact rational verification code, certificates, and the random area-selection record accompany the article.

Status. The counterexamples and classification are proved below. This is an AI-assisted, unrefereed research draft, not a proof-assistant-checked publication. A targeted literature search located the published open question but did not establish priority for the answer or for the classification. The conjugacy mechanism itself is not claimed to be new.

Part II (added 29 September 2026, batch 41; from Section 12 on) removes the single-circle restriction for increasing homeomorphisms of $\mathbb{R}$: it classifies the minimal iterative polynomials with arbitrary root moduli (Theorem 12.2), gives the exact universal order reduction $\mathcal{R}(P)$ of every equation with nonzero constant coefficient (Theorem 16.3), and decides affine rigidity and the existence of fixed points. It is likewise an AI-assisted, unrefereed draft with no priority claim.

Part III (added 29 September 2026, batch 42; from Section 25 on) classifies the minimal polynomials of decreasing homeomorphisms of $\mathbb{R}$ (Theorem 25.1), gives the exact universal reduction $\mathcal{C}_{\downarrow}(P)$, which needs one or two witnesses (Theorem 30.1), and proves polynomial-coordinate rigidity above a finite smoothness (Theorem 31.3). It is likewise AI-assisted and unrefereed, with no priority claim.

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1 The problem selected and the scope of the answer

1.1 A precise published target

Let $I \subseteq \mathbb{R}$ be a nondegenerate interval and let $f : I \rightarrow I$ be continuous. An equation

$$a_d f^{\circ d}(x) + a_{d-1} f^{\circ d-1}(x) + \cdots + a_1 f(x) + a_0 x = 0 \quad (1)$$

with $a_0 a_d \neq 0$ is a polynomial-like iterative equation. Its *characteristic polynomial* is $P(z) = \sum_{j=0}^d a_j z^j$. The notation concerns composition, not pointwise powers: in general, $f^{\circ 2}(x)$ is not $f(x)^2$.

The specific target is Problem 6.1 of Draga and Morawiec [1], in the last section of their paper. In the context of lowering the order of (1), they ask about deleting nonreal roots of largest modulus when a real root has that same modulus. The question also appears in the authors' preprint, page 13. Here *deleting roots* means replacing P by the corresponding polynomial divisor while preserving the implication that every solution satisfies the lower-order equation.

The answer under these hypotheses alone is negative. In fact, the cubic counterexample in Section 2 satisfies no nonzero equation of lower order whatsoever. The analytic counterexample in Section 3 shows that lack of smoothness is not the general reason for failure.

The equal-modulus qualification is essential. Existing order-reduction results employ strict separation of root moduli, together with orbit monotonicity or alternating monotonicity; Draga's later note discusses additional elimination cases and again identifies the unrestricted nonreal-root issue as unresolved [2]. Our examples do not contradict those theorems: they lie at the spectral boundary excluded by strict separation.

1.2 Relation to the initial candidate

The literature search began with Petrov's MathOverflow question asking for conditions on P that force a continuous solution of $P[f] = 0$ to be linear [3]. The retrieved question had no posted answer. It led to the more sharply formulated root-elimination problem above.

Throughout this article, *homogeneous linear* means $f(x) = cx$ and *affine* means $f(x) = cx + b$. The distinction matters when 1 is a multiple root. We do not claim a complete answer to Petrov's question for arbitrary polynomials or for both monotonicity orientations. We do obtain an exact answer for *increasing solutions when all characteristic roots lie on one circle*; see Section 8. [Added 29 September 2026, batch 41: Part II extends this answer to increasing solutions with arbitrary root moduli, Theorem 17.1.] [Added 29 September 2026, batch 42: Part III decides affine rigidity for decreasing solutions (Theorem 30.4); with Part II this answers Petrov's question for both orientations when $P(0) \neq 0$ (Remark 30.5).]

1.3 Random selection, exclusion, and priority

Before choosing a problem, a list of 80 mathematical areas was supplied to an operating-system-backed random generator through `secrets.randbelow(80)`. The first draw returned the zero-based index 17: area 18, *Functional equations*. There was no redraw. The complete ordered list and the result are preserved in `data/selection.json`.

The supplied `manifest(1).tex` was reviewed in full. The present problem was not selected from its entries. Its SHA-256 fingerprint is recorded with the selection so that the exclusion check can be tied to the actual supplied version; see Appendix B.

The target is therefore an explicitly published open question, not a new question invented for this report. The search found the original statement, subsequent related papers, and no explicit later answer. That is not an exhaustive bibliographic certification that the question remained open up to the preparation date. The mathematical conclusion below is independent of that priority issue: the proposed general elimination is false as printed. Nor should the elementary conjugacy construction be advertised as a new technique. The contribution developed here is the explicit counterexample, its connection with the published question, and the self-contained spectral classification.

2 An exact cubic counterexample

2.1 The two-branch definition

Define a function on one fundamental interval by

$$\phi(u) = \begin{cases} (u+5)/2, & 1 \leq u \leq 3, \\ 4u-8, & 3 \leq u \leq 8. \end{cases} \quad (2)$$

The formulas agree at $u = 3$. For $x > 0$, choose the unique $k \in \mathbb{Z}$ such that $8^k \leq x < 8^{k+1}$, and put

$$F(x) = 8^k \phi(8^{-k}x). \quad (3)$$

Complete the definition by

$$F(0) = 0, \quad F(-x) = -F(x) \quad (x > 0). \quad (4)$$

The endpoint identity $\phi(8) = 24 = 8\phi(1)$ makes the scaling construction consistent. It gives, on the whole real line,

$$F(8x) = 8F(x). \quad (5)$$

Theorem 2.1 (Cubic counterexample). *The function F defined by (2)–(4) is an odd, strictly increasing homeomorphism of $\mathbb{R}$ with*

$$\frac{1}{2}|x-y| \leq |F(x)-F(y)| \leq 4|x-y| \quad (x, y \in \mathbb{R}). \quad (6)$$

It satisfies

$$F^{\circ 3}(x) = 8x \quad (x \in \mathbb{R}), \quad (7)$$

but is not homogeneous linear or affine. The smallest degree of a nonzero polynomial Q with $Q[F] = 0$ is three, and the monic polynomial of that degree is

$$\mu_F(z) = z^3 - 8 = (z-2)(z^2 + 2z + 4). \quad (8)$$

Proof. On every scaled copy of the two intervals in (2), the slope of F is either $1/2$ or 4 . The pieces meet continuously. Moreover,

$$\frac{4}{3} \leq \frac{F(x)}{x} \leq 3 \quad (x > 0). \quad (9)$$

For the first branch this follows from $\phi(u)/u = 1/2 + 5/(2u)$; for the second it follows from $\phi(u)/u = 4 - 8/u$. Thus $F(x) \rightarrow 0$ as $x \rightarrow 0$, and $F(x) \rightarrow \infty$ as $x \rightarrow \infty$. Oddness handles the negative half-line. Strict monotonicity and surjectivity follow.

Between two strictly positive points there are only finitely many breakpoints, so their secant slope lies between $1/2$ and 4 . Pass to a limit when one endpoint is zero, and use oddness and addition across zero to obtain (6) for all pairs of points.

To check the iterate relation it suffices to work on $[1, 8]$. There are three relevant itinerary intervals, because an intermediate iterate crosses a breakpoint at $x = 4$. Direct substitution gives Table 1. In using the table, values in $[8, 64]$ are computed with (5). The final column is $8x$ in every row. Scaling extends the identity to every positive x ; oddness and the value at zero finish the proof of (7).

The values $F(1) = 3$ and $F(3) = 4$ show that F is not homogeneous linear. Since $F(0) = 0$, it is not affine either. Minimality is proved immediately below. $\square$

Table 1: Exact branchwise verification of the cubic relation.

Initial interval	$F(x)$	$F^{\circ 2}(x)$	$F^{\circ 3}(x)$
$[1, 3]$	$(x + 5)/2$	$2x + 2$	$8x$
$[3, 4]$	$4x - 8$	$16x - 40$	$8x$
$[4, 8]$	$4x - 8$	$2x + 16$	$8x$

2.2 A small exact certificate of minimality

The orbit beginning at 1 is

$$1, 3, 4, 8, 24, 32, 64, 192, 256, \dots \quad (10)$$

In particular,

$$\det \begin{pmatrix} 1 & 3 & 4 \\ 3 & 4 & 8 \\ 4 & 8 & 24 \end{pmatrix} = -56 \neq 0. \quad (11)$$

Suppose $q_0x + q_1F(x) + q_2F^{\circ 2}(x) = 0$ on $\mathbb{R}$. Substitution of $x = 1, F(1), F^{\circ 2}(1)$ gives the matrix in (11) times $(q_0, q_1, q_2)^T$ equal to zero. Hence all three coefficients vanish. This rules out every nonzero relation of degree at most two. Relation (7) supplies a monic relation of degree three. Subtracting any other monic cubic relation would produce a relation of lower degree, so (8) is unique.

The roots of μ_F are

$$2, \quad -1 + i\sqrt{3}, \quad -1 - i\sqrt{3}. \quad (12)$$

They all have modulus 2. Deleting the nonreal pair leaves the polynomial $z - 2$, but $(z - 2)[F](1) = 1 \neq 0$. This is precisely the obstruction asked about in the selected problem. Degree three is also the smallest possible degree for a real polynomial to have both a real root and a nonreal conjugate pair.

2.3 The conjugacy behind the formula

The construction can also be understood without the itinerary table. On $[1, 8]$ define

$$h_0(u) = \begin{cases} 2u - 1, & 1 \leq u \leq 2, \\ u/2 + 2, & 2 \leq u \leq 4, \\ u, & 4 \leq u \leq 8. \end{cases} \quad (13)$$

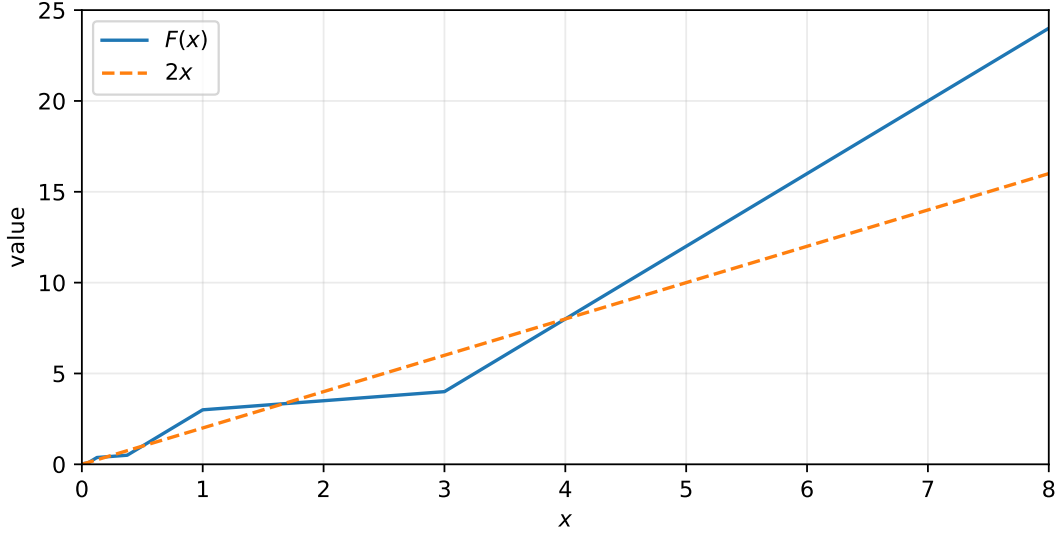


Figure 1: The exact cubic counterexample on one scale interval, compared with the homogeneous linear solution. There are infinitely many scaled copies of the two affine branches, accumulating at zero.

Extend it by $H(8^k u) = 8^k h_0(u)$ for $u \in [1, 8]$, then extend oddly and set $H(0) = 0$. The map H is an increasing homeomorphism, and it commutes with $D_8(x) = 8x$. Direct calculation shows

$$F = H \circ D_2 \circ H^{-1}. \quad (14)$$

Consequently,

$$F^{\circ 3} = H \circ D_8 \circ H^{-1} = D_8. \quad (15)$$

The middle equality is useful only because H commutes with D_8 ; it need not commute with D_2 .

There is a general warning here. Arbitrary nonlinear conjugacy does *not* preserve an arbitrary linear combination of iterates. Although iterates conjugate one by one, a nonlinear conjugating map cannot be moved through a sum. In (14), commutation with the specific third iterate supplies the extra fact needed. In later constructions, we impose a linear difference equation on the conjugating coordinate itself instead.

2.4 A visible failure of differentiability

The scaled sequences $x_n = 8^{-n}$ and $y_n = 3 \cdot 8^{-n}$ both tend to zero, but

$$\frac{F(x_n)}{x_n} = 3, \quad \frac{F(y_n)}{y_n} = \frac{4}{3}. \quad (16)$$

Thus F is not differentiable at zero, despite being globally bi-Lipschitz. Section 9 explains why this is inevitable for a square-free circular spectrum of radius different from 1. The next example avoids this regularity issue altogether.

3 A real-analytic counterexample on the entire real line

Let

$$h(t) = t + \frac{1}{4} \sin \frac{\pi t}{2}, \quad f(x) = h(h^{-1}(x) + 1). \quad (17)$$

Here h^{-1} is the ordinary inverse function, not a reciprocal. Since

$$h'(t) = 1 + \frac{\pi}{8} \cos \frac{\pi t}{2} \quad \text{and} \quad 1 - \frac{\pi}{8} > 0, \quad (18)$$

h is strictly increasing. Also $|h(t) - t| \leq 1/4$, so it maps $\mathbb{R}$ onto $\mathbb{R}$. Its derivative never vanishes. The real-analytic inverse function theorem, applied at each real point, makes h^{-1} and f real analytic. No global complex-analytic inverse is asserted or required.

Theorem 3.1 (Analytic quartic counterexample). *The function in (17) is an increasing, globally bi-Lipschitz, real-analytic homeomorphism of $\mathbb{R}$. Its minimal iterative polynomial is*

$$\mu_f(z) = (z - 1)^2(z^2 + 1) = z^4 - 2z^3 + 2z^2 - 2z + 1. \quad (19)$$

It also satisfies $f^{\circ 4}(x) = x + 4$. The equation obtained by deleting the nonreal roots from (19) is not satisfied.

Proof. Conjugacy gives, for every integer n ,

$$f^{\circ n}(h(t)) = h(t + n) = t + n + \frac{1}{4} \sin \frac{\pi(t + n)}{2}. \quad (20)$$

The linear polynomial in n is killed by $(E - 1)^2$, where E shifts n to $n + 1$. The sine term is killed by $E^2 + 1$. The two operators commute, so their product kills the entire sequence. Therefore $\mu_f[f] = 0$. Periodicity of the sine also gives $h(t + 4) = h(t) + 4$, proving $f^{\circ 4}(x) = x + 4$.

The first orbit values at zero are

$$0, \quad \frac{5}{4}, \quad 2, \quad \frac{11}{4}, \quad 4, \quad \frac{21}{4}, \quad 6, \dots \quad (21)$$

Their 4×4 Hankel determinant is

$$\det \begin{pmatrix} 0 & 5/4 & 2 & 11/4 \\ 5/4 & 2 & 11/4 & 4 \\ 2 & 11/4 & 4 & 21/4 \\ 11/4 & 4 & 21/4 & 6 \end{pmatrix} = 1. \quad (22)$$

The same argument used for (11) rules out every relation of degree below four. This proves minimality.

After deleting i and $-i$, the remaining polynomial is $(z - 1)^2$. At $x = 0$ its residual is

$$f^{\circ 2}(0) - 2f(0) + 0 = 2 - 2 \cdot \frac{5}{4} = -\frac{1}{2}. \quad (23)$$

Thus the proposed deletion fails. Finally, $f'(h(t)) = h'(t + 1)/h'(t)$ is bounded above and bounded away from zero, which proves the bi-Lipschitz assertion. $\square$

All roots in (19) lie on the unit circle, and 1 is a real root of maximal modulus. This gives a second negative answer, now with real analyticity at every real point and with all polynomial coefficients nonzero. Notice that the map has no fixed point: $h(t + 1) > h(t)$. The distinction between a fixed point and a translation-like orbit will account for the different multiplicity rules at radius 1.

4 Orbit recurrences and minimal iterative polynomials

We now develop the tools for a general classification. The results in this section also explain why an exact orbit determinant is a valid minimality certificate.

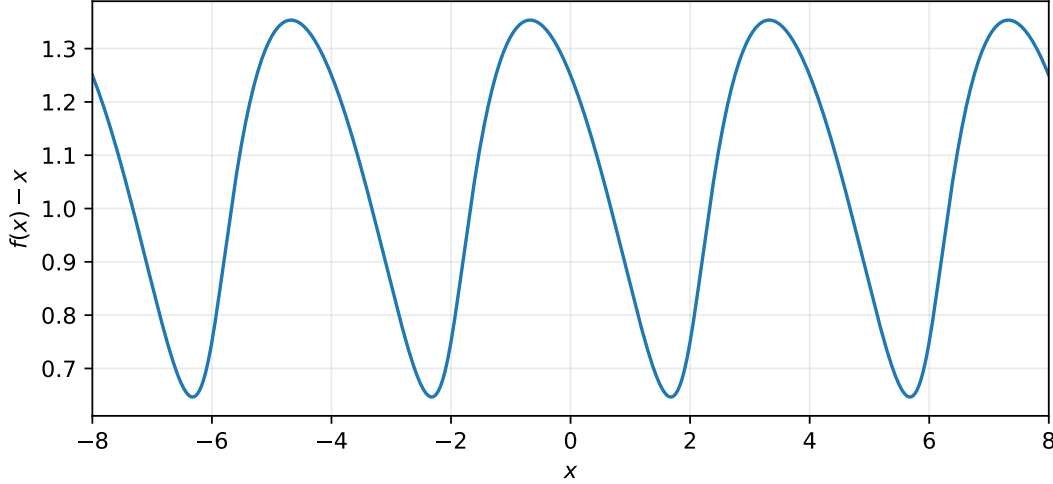


Figure 2: Numerical illustration of $f(x) - x$ for (17). The displacement is nonconstant and 4-periodic. The identities and the counterexample are established exactly in the text, not by this plot.

4.1 The annihilator ideal

Definition 4.1. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, let

$$\text{Ann}(f) = \{P \in \mathbb{R}[z] : P[f](x) = 0 \text{ for every } x \in \mathbb{R}\}. \quad (24)$$

If this set contains a nonzero polynomial, its monic polynomial of least degree is called the *minimal iterative polynomial* and is denoted by μ_f .

Lemma 4.2. *The set $\text{Ann}(f)$ is an ideal of $\mathbb{R}[z]$. When it is nonzero, $\text{Ann}(f) = (\mu_f)$.*

Proof. Closure under addition and scalar multiplication is immediate. For every polynomial P ,

$$(zP)[f](x) = P[f](f(x)). \quad (25)$$

Thus $P \in \text{Ann}(f)$ implies $zP \in \text{Ann}(f)$, and consequently every polynomial multiple of P belongs to the ideal. Divide any annihilator by the monic least-degree one. The remainder is another annihilator and has smaller degree, so it must be zero. $\square$

It is *right composition* in (25) that respects linear combinations. Applying f to a linear combination of its iterates would not give the same conclusion.

4.2 Why both forward and backward iterates exist

Lemma 4.3. *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $P[f] = 0$ for a polynomial with $P(0) \neq 0$, then f is a strictly monotone homeomorphism of $\mathbb{R}$.*

Proof. If $f(x) = f(y)$, then every positive iterate has the same value at x and y . Subtracting the two equations $P[f](x) = P[f](y) = 0$ gives $P(0)(x - y) = 0$, hence $x = y$. A continuous injective real function is strictly monotone.

If its image were a proper interval, at least one of its limits at $+\infty$ and $-\infty$ would be finite. Along the corresponding end, $f(x)$ would tend to a finite number, and continuity would make every positive iterate tend to a finite number as well. But the term $P(0)x$ in the equation would be unbounded. This contradiction proves surjectivity. A continuous strictly monotone bijection of $\mathbb{R}$ has a continuous inverse. $\square$

For such an f , define f^{on} for every $n \in \mathbb{Z}$ using inverse iterates when $n < 0$. For any starting point x , its orbit $u_x(n) = f^{on}(x)$ satisfies

$$\sum_{j=0}^d a_j u_x(n+j) = 0 \quad (n \in \mathbb{Z}). \quad (26)$$

Indeed, substitute $f^{on}(x)$ into $P[f] = 0$.

4.3 Exponential-polynomial solutions

Lemma 4.4 (Two-sided recurrence decomposition). *Let P be a polynomial with nonzero leading and constant coefficients. Write its distinct complex roots as λ , with multiplicities m_λ . Every complex sequence $u : \mathbb{Z} \rightarrow \mathbb{C}$ satisfying $P(E)u = 0$ has a unique representation*

$$u(n) = \sum_{\lambda} p_{\lambda}(n) \lambda^n, \quad \deg p_{\lambda} < m_{\lambda}. \quad (27)$$

For a real sequence, conjugate roots have conjugate coefficient polynomials. Its minimal recurrence polynomial is

$$\prod_{p_{\lambda} \neq 0} (z - \lambda)^{1 + \deg p_{\lambda}}. \quad (28)$$

Proof. For a polynomial p ,

$$(E - \mu)(p(n)\lambda^n) = \lambda^n(\lambda p(n+1) - \mu p(n)). \quad (29)$$

When $\mu = \lambda$, a nonconstant polynomial degree drops by one, and a constant is killed. When $\mu \neq \lambda$, the degree is preserved and the leading coefficient is multiplied by $\lambda - \mu$.

These observations prove that the proposed sequences satisfy the recurrence. They also prove their linear independence: in any putative linear relation, apply enough factors corresponding to all roots except one. The surviving root component is a nonzero polynomial times an exponential unless its coefficient polynomial was zero. There are $\deg P$ independent sequences. A two-sided recurrence with nonzero first and last coefficients is determined by $\deg P$ consecutive values, so these sequences form a basis. Formula (29) then proves the exact multiplicities in (28). $\square$

Lemma 4.5 (From individual orbits to the whole function). *The minimal iterative polynomial is the least common multiple of the minimal recurrence polynomials of all scalar orbits u_x . In particular, for each root, its multiplicity in μ_f is the maximum of the multiplicities with which it occurs in the scalar orbits.*

Proof. Each orbit is annihilated by μ_f , so its minimal recurrence polynomial divides μ_f . Conversely, any common multiple of all the orbit polynomials annihilates every orbit at $n = 0$, and hence annihilates f . Only finitely many roots and bounded multiplicities are possible, because they all occur in μ_f . Therefore their least common multiple and each stated maximum are attained using finitely many orbits. $\square$

This also gives a finite-dimensional interpretation. Right composition by f acts linearly on the span of $\text{id}, f, f^{\circ 2}, \dots$. The vector id is cyclic, and μ_f is the minimal polynomial on this cyclic space. The function f itself need not be linear as a map of real numbers.

5 The positivity lemma that controls a circular spectrum

A finite exponential polynomial on the unit circle is a sequence

$$v(n) = \sum_{\zeta \in S} p_{\zeta}(n) \zeta^n, \quad n \in \mathbb{Z}, \quad (30)$$

where S is a finite set of distinct complex numbers of modulus one. The sequence is real when the terms respect conjugation. The root $\zeta = 1$ contributes the nonoscillating polynomial part.

5.1 A recurrence fact for finite trigonometric sequences

Lemma 5.1. *For any finite set S on the unit circle, there are positive integers $N_k \rightarrow \infty$ such that $\zeta^{N_k} \rightarrow 1$ simultaneously for every $\zeta \in S$. Consequently, for a finite trigonometric sequence $T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n$ and any fixed $\ell \in \mathbb{Z}$,*

$$T(N_k + \ell) \rightarrow T(\ell), \quad T(-N_k + \ell) \rightarrow T(\ell). \quad (31)$$

Proof. The vectors $(\zeta^n)_{\zeta \in S}$ lie in a compact finite product of circles. Choose a convergent subsequence with indices n_k whose successive gaps tend to infinity. The componentwise quotient of the vectors at indices n_{2k} and n_k tends to the identity vector. Set $N_k = n_{2k} - n_k$. The inverse quotients converge to the identity as well, giving both assertions in (31). $\square$

Lemma 5.2 (Nonnegative trigonometric sequences have positive mean). *Let $T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n$ be real, nonnegative for every $n \in \mathbb{Z}$, and not identically zero. Then $1 \in S$ and $c_1 > 0$.*

Proof. The Cesàro mean of ζ^n over $n = 0, \dots, N-1$ tends to zero for $\zeta \neq 1$, by the finite geometric-sum formula. Thus the mean of T tends to c_1 , interpreted as zero if $1 \notin S$. Distinctness of the frequencies gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |T(n)|^2 = \sum_{\zeta \in S} |c_{\zeta}|^2 > 0. \quad (32)$$

On the other hand, $0 \leq T(n) \leq M$ for some finite M , and $T(n)^2 \leq MT(n)$. Taking means proves $c_1 > 0$. $\square$

5.2 Parity and dominance of the nonoscillating term

Theorem 5.3 (Two-sided positivity). *Suppose the real sequence (30) is nonnegative for all $n \in \mathbb{Z}$ and is not identically zero. Let $d = \max\{\deg p_{\zeta} : p_{\zeta} \neq 0\}$. Then d is even. The polynomial p_1 is present, has degree exactly d , and has positive leading coefficient. The analogous statement for a nonpositive sequence holds after multiplication by -1 .*

Proof. Let c_{ζ} be the coefficient of n^d in p_{ζ} , with zero used when its degree is lower, and set

$$T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n. \quad (33)$$

The leading trigonometric sequence is not identically zero, by the independence in Lemma 4.4. Uniformly as $|n| \rightarrow \infty$,

$$v(n) = n^d T(n) + O(|n|^{d-1}), \quad (34)$$

with the evident exact interpretation when $d = 0$.

At $+\infty$, nonnegativity gives $\liminf T(n) \geq 0$. Use the first sequence in (31) to conclude $T(\ell) \geq 0$ for every fixed integer ℓ . At $-\infty$, division by $|n|^d$ and the second return sequence give $(-1)^d T(\ell) \geq 0$ for every ℓ . If d were odd, T would vanish identically, a contradiction. Thus d is even. Lemma 5.2 now gives $c_1 > 0$. Hence p_1 has degree d and the asserted leading sign. $\square$

The use of both time directions is decisive. For a recurrence constrained only at nonnegative indices, an odd-degree polynomial may remain positive. A homeomorphism supplies all negative iterates, eliminating that possibility.

Corollary 5.4 (A vanishing unit-circle exponential polynomial). *If an exponential polynomial (30) tends to zero as $n \rightarrow +\infty$ or as $n \rightarrow -\infty$, it is identically zero.*

Proof. Choose its largest polynomial degree d and its leading trigonometric sequence T as in (34). The assumed limit implies $T(n) \rightarrow 0$ in the chosen direction. The corresponding return sequence in Lemma 5.1 gives $T(\ell) = 0$ for every ℓ , contrary to the choice of a nonzero leading degree. Thus a nonzero exponential polynomial cannot have the stated limit. $\square$

6 Classification of minimal polynomials on one circle

Call a polynomial *circular of radius r* if every one of its complex roots has modulus r . Multiplicities are counted in the usual algebraic sense. A missing root is assigned multiplicity zero.

Theorem 6.1 (Circular-spectrum classification). *Let $Q \in \mathbb{R}[z]$ be nonconstant and monic, and suppose all its roots lie on $|z| = r$ for some $r > 0$. Then Q is the minimal iterative polynomial of an increasing homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$ if and only if the following conditions hold.*

- (i) *If $r \neq 1$, the positive real number r is a root of odd multiplicity m_r , and every other root has multiplicity at most m_r .*
- (ii) *If $r = 1$, either $Q(z) = z - 1$, or 1 is a root of even multiplicity $m_1 \geq 2$ and every other root has multiplicity at most $m_1 - 1$.*

In case (i), realizations can be chosen odd, globally bi-Lipschitz, and real analytic away from zero. In the nonidentity part of case (ii), realizations can be chosen globally bi-Lipschitz and real analytic at every real point; they have no fixed point.

This is a classification of the polynomials that can be *minimal*. It is not a parametrization of every function realizing any one polynomial. The minimality qualification is essential: for example, the identity satisfies $(z - 1)^7 [\text{id}] = 0$, but its minimal polynomial is only $z - 1$. [Added 29 September 2026, batch 41: Theorem 12.2 of Part II classifies the minimal polynomials with roots of arbitrary moduli; on one circle it reduces to the present theorem (Remark 15.3).]

6.1 Necessity when $r \neq 1$

Let f be increasing and let $\mu_f = Q$. By Lemma 4.4, every orbit has the form

$$u_x(n) = r^n \sum_{|\zeta|=1} p_{x,\zeta}(n) \zeta^n. \quad (35)$$

If $r > 1$, this tends to zero as $n \rightarrow -\infty$; if $r < 1$, it tends to zero as $n \rightarrow +\infty$. Passing to this limit in $f(u_x(n)) = u_x(n+1)$ gives $f(0) = 0$. Since f and f^{-1} are increasing and fix zero, every nonzero orbit keeps the sign of its starting point for all $n \in \mathbb{Z}$.

For $x > 0$, divide (35) by $r^n > 0$ and apply Theorem 5.3. Its largest polynomial degree is even and is attained by the coefficient of $\zeta = 1$, corresponding to the characteristic root r . The same conclusion holds for $x < 0$ after a sign change. Thus, in every nonzero scalar orbit, the multiplicity of r is odd and is at least as large as every other multiplicity.

Now take the maxima over orbits using Lemma 4.5. The maximum of a nonempty finite set of odd multiplicities is odd. Every other global multiplicity is bounded by that maximum. This proves the necessity of condition (i).

6.2 Necessity on the unit circle

Suppose $r = 1$ and $f \neq \text{id}$. Choose a nonfixed starting point x . Because f is increasing, the difference sequence

$$v_x(n) = u_x(n+1) - u_x(n) \quad (36)$$

has a strict, constant sign for all $n \in \mathbb{Z}$.

Taking a first difference lowers the degree of the polynomial component at the root 1 by one. At every other root it preserves the degree, as (29) shows. Apply Theorem 5.3 to v_x , after a sign change if necessary. Its largest degree is even and is attained at 1. Therefore the polynomial component of u_x at 1 has odd degree $d \geq 1$, and every oscillatory component has degree at most $d - 1$. Equivalently, this orbit has an even multiplicity $d + 1$ at 1, strictly larger than all the other multiplicities.

The nonoscillating leading term dominates the orbit at both ends:

$$u_x(n) = an^d + O(|n|^{d-1}), \quad a \neq 0, \quad d \text{ odd}. \quad (37)$$

In particular, the two time tails go to opposite infinities. If f had a fixed point, an increasing orbit starting on either side of it could never cross it, contradicting (37). Hence a nonidentity f in this setting has no fixed point, and all its orbits are nonfixed. Lemma 4.5 now gives an even maximal multiplicity at 1 and the strict bound on every other multiplicity. If instead $f = \text{id}$, the minimal polynomial is $z - 1$. This proves condition (ii).

6.3 Encoding the remaining root directions

For the construction, write every root other than the distinguished positive root using an angle $0 < \theta_j \leq \pi$. An angle $0 < \theta_j < \pi$ denotes the conjugate pair $re^{i\theta_j}, re^{-i\theta_j}$, with common multiplicity m_j . The angle $\theta_j = \pi$ denotes the single real root $-r$, if present, with multiplicity m_j . Choose distinct angles and set

$$e_j = m_j - 1. \quad (38)$$

A term $r^{e_j} \cos(\theta_j t)$ then produces precisely the desired degree in its sampled orbit. At angle π , the two exponential expressions for the cosine coincide on integer samples and give the one real root, not two roots.

6.4 Construction for $r \neq 1$

Write $m_r = 2k + 1$ and set $a = \log r \neq 0$. Choose $C \geq 1$ so large that

$$\frac{k}{\sqrt{C}} \leq \frac{|a|}{2}, \quad (39)$$

and define

$$A(t) = (t^2 + C)^k, \quad W(t) = A(t) + \sum_j \varepsilon_j t^{e_j} \cos(\theta_j t), \quad h(t) = r^t W(t), \quad (40)$$

where every ε_j is positive and sufficiently small. The permitted multiplicities give $0 \leq e_j \leq 2k$. An explicit sufficient choice is any collection satisfying

$$\sum_j \varepsilon_j (|a| + e_j + \theta_j) \leq \frac{|a|}{4}. \quad (41)$$

When there are no other roots, the sum is empty.

Here are the estimates that justify the construction. For $0 \leq e \leq 2k$,

$$|t|^e \leq A(t), \quad e|t|^{e-1} \leq eA(t) \quad (e \geq 1), \quad \frac{|A'(t)|}{A(t)} \leq \frac{k}{\sqrt{C}}. \quad (42)$$

The first inequalities follow separately on $|t| \leq 1$ and $|t| \geq 1$. The last one follows by maximizing $2k|t|/(t^2 + C)$. Since (41) implies $\sum_j \varepsilon_j \leq 1/4$,

$$\frac{3}{4}A(t) \leq W(t) \leq \frac{5}{4}A(t). \quad (43)$$

Also

$$h'(t) = r^t (aW(t) + W'(t)). \quad (44)$$

The unperturbed expression $aA + A'$ has the sign of a and absolute value at least $|a|A/2$. Each perturbing expression

$$at^{e_j} \cos(\theta_j t) + e_j t^{e_j-1} \cos(\theta_j t) - \theta_j t^{e_j} \sin(\theta_j t) \quad (45)$$

has absolute value at most $(|a| + e_j + \theta_j)A(t)$, with the middle term omitted for $e_j = 0$. Thus

$$\operatorname{sgn} h'(t) = \operatorname{sgn} a, \quad \frac{|a|}{4} r^t A(t) \leq |h'(t)| \leq \frac{7|a|}{4} r^t A(t). \quad (46)$$

It follows that h is a real-analytic bijection from $\mathbb{R}$ onto $(0, \infty)$: it is increasing for $r > 1$ and decreasing for $r < 1$. Define

$$f(x) = h(h^{-1}(x) + 1) \quad (x > 0), \quad f(0) = 0, \quad f(-x) = -f(x). \quad (47)$$

Even when h is decreasing, the composition in (47) is increasing. Its limit at zero is zero, so the odd extension is an increasing homeomorphism of $\mathbb{R}$.

Every orbit on the positive half-line is $h(t + n)$. Expanding its cosine terms shows that it is annihilated by Q . Oddness handles negative orbits. More importantly, at the starting point $h(0)$ the sampled sequence is

$$h(n) = r^n \left((n^2 + C)^k + \sum_j \varepsilon_j n^{e_j} \cos(\theta_j n) \right). \quad (48)$$

The component at r has degree exactly $2k$, and every other component has exactly its assigned degree e_j , with nonzero leading coefficient. Lemma 4.4 proves that this single orbit has minimal polynomial Q . Hence $\mu_f = Q$.

Finally, (46) bounds $f'(h(t)) = h'(t+1)/h'(t)$ above and away from zero, because $A(t+1)/A(t)$ is positive, continuous, and tends to 1 at both ends. The resulting derivative bounds extend to secant bounds across zero. This proves the bi-Lipschitz and regularity assertions in case (i).

6.5 Construction on the unit circle

For the identity case choose $f = \text{id}$. Otherwise write $m_1 = 2k + 2$ and put

$$A(t) = \int_0^t (1 + s^2)^k ds = \sum_{\ell=0}^k \binom{k}{\ell} \frac{t^{2\ell+1}}{2\ell+1}. \quad (49)$$

Now A is an odd polynomial of degree $2k + 1$ and $A'(t) = (1 + t^2)^k > 0$. Since $e_j \leq 2k$, set

$$h(t) = A(t) + \sum_j \varepsilon_j t^{e_j} \cos(\theta_j t) \quad (50)$$

with all $\varepsilon_j > 0$ chosen so that

$$\sum_j \varepsilon_j (e_j + \theta_j) < \frac{1}{2}. \quad (51)$$

The same estimates as in (42), now relative to A' , imply

$$\frac{1}{2}(1 + t^2)^k < h'(t) < \frac{3}{2}(1 + t^2)^k. \quad (52)$$

The polynomial A has degree one more than every perturbing amplitude, so $h(t) \rightarrow \pm\infty$ as $t \rightarrow \pm\infty$. Therefore h is a real-analytic increasing diffeomorphism of $\mathbb{R}$.

Set $f = h \circ T \circ h^{-1}$, where $T(t) = t + 1$. All its orbits are $h(t + n)$ and are annihilated by Q . The orbit $h(n)$ contains a nonoscillating polynomial of degree $2k + 1$ and exactly the prescribed oscillating polynomial degrees e_j . Thus its minimal recurrence polynomial, and hence μ_f , is Q . The derivative ratio from (52) is bounded above and away from zero, proving global bi-Lipschitz continuity. The conjugacy also gives $f(x) > x$ everywhere and therefore no fixed point. This completes the sufficiency proof and the proof of Theorem 6.1.

7 Quantitative square-free realizations

The simplest part of the classification has a particularly transparent formula. It shows that survival of nonreal roots is not restricted to roots of unity or to an isolated cubic.

Fix $r > 0$, $r \neq 1$, and distinct angles $0 < \theta_j \leq \pi$. Choose nonzero real amplitudes ε_j and put

$$W(t) = 1 + \sum_{j=1}^q \varepsilon_j \cos(\theta_j t), \quad h(t) = r^t W(t). \quad (53)$$

Let $a = \log r$ and

$$B = \sum_{j=1}^q |\varepsilon_j| \sqrt{a^2 + \theta_j^2}. \quad (54)$$

Proposition 7.1. *If $B < |a|$, then the odd extension of $f = h \circ T \circ h^{-1}$ from $(0, \infty)$ to $\mathbb{R}$ is an increasing homeomorphism with minimal polynomial*

$$(z - r) \prod_{0 < \theta_j < \pi} (z^2 - 2r \cos \theta_j z + r^2) \prod_{\theta_j = \pi} (z + r). \quad (55)$$

It is real analytic away from zero, and its positive derivative on either open half-line satisfies

$$r \frac{|a| - B}{|a| + B} \leq f'(x) \leq r \frac{|a| + B}{|a| - B}. \quad (56)$$

These are also valid global lower and upper Lipschitz constants.

Proof. Since $B \geq |a| \sum_j |\varepsilon_j|$, the condition gives $W(t) > 0$. Moreover,

$$aW(t) + W'(t) = a + \sum_j \varepsilon_j (a \cos(\theta_j t) - \theta_j \sin(\theta_j t)) \quad (57)$$

differs from a by at most B . Thus h' has the sign of a , and its absolute value lies between $r^t(|a| - B)$ and $r^t(|a| + B)$. This proves that h is a suitable coordinate, and taking the ratio at $t + 1$ and t gives (56). Minimality follows from the nonzero frequency components in $h(n)$, exactly as before. $\square$

For example, taking one angle $0 < \theta < \pi$ yields a nonlinear solution whose minimal polynomial is

$$(z - r)(z^2 - 2r \cos \theta z + r^2). \quad (58)$$

The angle can be any such real number, not necessarily a rational multiple of π . In particular, $\theta = \pi/2$ and $r = 2$ give the cubic $z^3 - 2z^2 + 4z - 8$, all of whose coefficients are nonzero.

A small oscillatory component can coexist with strict monotonicity because the exponential envelope supplies a nonzero derivative margin. It is incorrect to infer that an orbit must lose every oscillatory mode merely because the unnormalized orbit is monotone. The cubic orbit (10) makes this visible:

$$\frac{F^{on}(1)}{2^n} = 1, \frac{3}{2}, 1, 1, \frac{3}{2}, 1, \dots \quad (59)$$

The normalized values oscillate, while the actual orbit is strictly increasing.

8 Sharp rigidity consequences for increasing solutions

The classification concerns minimal polynomials, but a given equation need not be minimal. Passing to divisors gives explicit rigidity tests.

Corollary 8.1 (Nonunit circle). *Let P be a nonconstant monic real polynomial whose roots lie on $|z| = r \neq 1$. If r is not a root, there is no increasing continuous solution of $P[f] = 0$ on $\mathbb{R}$. If r is a root, every increasing continuous solution is $f(x) = rx$ if and only if*

$$P(z) = (z - r)^m \quad \text{with } m \in \{1, 2\}. \quad (60)$$

Proof. Any such continuous solution is a homeomorphism by Lemma 4.3, and its minimal polynomial divides P . Theorem 6.1 requires the positive root r . Under (60), the only permitted odd multiplicity is one, so $\mu_f = z - r$.

Conversely, if P has any other root, take that real negative root or that nonreal conjugate pair together with a simple root at r . The square-free construction realizes the resulting divisor of P as a minimal polynomial of degree at least two; the realization is not homogeneous linear. If P has no other root but $m \geq 3$, the construction with $\mu_f = (z - r)^3$ supplies a nonlinear solution. Every solution in this nonunit setting fixes zero, so affine and homogeneous-linear rigidity coincide. $\square$

Corollary 8.2 (Unit circle). *Let P be nonconstant, monic, real, and supported on $|z| = 1$. Let M be the multiplicity of 1 in P .*

- (i) *If $M = 0$, there is no increasing continuous solution on $\mathbb{R}$.*
- (ii) *If $M \geq 1$, every increasing continuous solution is the identity if and only if $M = 1$.*
- (iii) *Among equations with an increasing solution, every increasing continuous solution is affine if and only if either $M = 1$, or $P(z) = (z - 1)^M$ with $M = 2$ or $M = 3$.*

Proof. If $M = 0$, the classification rules out every minimal divisor. If $M = 1$, the only possible minimal divisor is $z - 1$, regardless of the multiplicities of the other roots. When $M \geq 2$, every translation $x \mapsto x + c$ satisfies the equation, because its minimal polynomial for $c \neq 0$ is $(z - 1)^2$. This proves (i) and (ii).

If $P = (z - 1)^M$ with $M = 2$ or 3 , the only nonidentity minimal divisor allowed by the classification is $(z - 1)^2$. To see directly that its solutions are translations, write an orbit as $f^{cn}(x) = x + n(f(x) - x)$. For $x < y$, order preservation for every $n \in \mathbb{Z}$ gives

$$(y - x) + n((f(y) - y) - (f(x) - x)) > 0. \quad (61)$$

The coefficient of n must be zero, so $f(x) - x$ is constant.

If $M \geq 4$, a nonlinear realization of $(z - 1)^4$ is obtained from $h(t) = t + t^3/3$. If $M \geq 2$ and P has another root, adjoining that negative root or nonreal conjugate pair to $(z - 1)^2$ gives an admissible minimal divisor of degree greater than two. Its realization cannot be affine. Indeed, an affine increasing map with slope different from one has that positive slope as a characteristic root, whereas the only positive root here is one; an affine map of slope one is a translation and has minimal degree at most two. $\square$

These criteria make the radius-1 boundary especially clear. A simple positive root at 1 eliminates all other circular modes for increasing solutions. A double root at 1 supplies enough linear drift to allow nonconstant bounded oscillations. The analytic quartic example realizes exactly that first nonreal possibility. [Added 29 September 2026, batch 41: for arbitrary root moduli, affine and homogeneous-linear rigidity are decided by the universal core $\mathcal{R}(P)$ of Part II, Theorem 17.1; on one circle this gives the two corollaries above (Remark 17.2).]

9 What additional regularity does and does not restore

9.1 Differentiability is enough in the square-free nonunit case

Theorem 9.1. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and continuous, and suppose its minimal iterative polynomial is square-free and supported on $|z| = r \neq 1$. If f is differentiable at zero, then $f(x) = rx$ for every x .*

Proof. We already know that $f(0) = 0$. Write $L = f'(0)$. Differentiating the iterative relation at zero, using the chain rule and the fixed point, gives $\mu_f(L) = 0$. Since an increasing differentiable map has $L \geq 0$, and zero is not a root, the only possible value is $L = r$.

For a fixed starting point, square-freeness gives $u(n) = r^n T(n)$, where T is a bounded finite trigonometric sequence. In the time direction in which $u(n) \rightarrow 0$, differentiability implies

$$u(n+1) - ru(n) = o(|u(n)|). \quad (62)$$

Divide by r^{n+1} and use boundedness of T to obtain $T(n+1) - T(n) \rightarrow 0$. This difference is a unit-circle exponential polynomial, so Corollary 5.4 makes it identically zero. In particular, $u(1) = ru(0)$. Since the starting point was arbitrary, $f = r \text{ id}$. $\square$

9.2 A positive-order Taylor remainder handles repeated roots

Theorem 9.2. *Let f be increasing and continuous, with a minimal iterative polynomial supported on $|z| = r \neq 1$, allowing repeated roots. Suppose f is differentiable at zero and, for some $\alpha > 0$,*

$$f(x) = f'(0)x + O(|x|^{1+\alpha}) \quad (x \rightarrow 0). \quad (63)$$

Then $f(x) = rx$ on $\mathbb{R}$. In particular, local $C^{1,\alpha}$ regularity at zero is sufficient.

Proof. As above, $f(0) = 0$ and $f'(0) = r$. Each orbit has the form $u(n) = r^n v(n)$, where v is a unit-circle exponential polynomial and $|v(n)| = O((1+|n|)^d)$ for some d . Apply (63) in the time direction in which $r^n \rightarrow 0$. It gives

$$v(n+1) - v(n) = \frac{u(n+1) - ru(n)}{r^{n+1}} \quad (64)$$

$$= O\left(r^{\alpha n} (1+|n|)^{d(1+\alpha)}\right) \rightarrow 0. \quad (65)$$

The last limit holds both for $r < 1, n \rightarrow +\infty$ and for $r > 1, n \rightarrow -\infty$. By Corollary 5.4, the difference vanishes identically. Thus $u(1) = ru(0)$ for every starting point. $\square$

9.3 C^1 alone is not sufficient with repeated roots

Consider the particularly simple coordinate

$$h(t) = e^t(t^2 + 4), \quad h'(t) = e^t((t+1)^2 + 3) > 0. \quad (66)$$

Use $f = h \circ T \circ h^{-1}$ on $(0, \infty)$ and extend it oddly through zero. Its minimal iterative polynomial is $(z - e)^3$. For $x = h(t)$,

$$\frac{f(x)}{x} = e \frac{(t+1)^2 + 4}{t^2 + 4} \rightarrow e \quad (t \rightarrow -\infty). \quad (67)$$

Also $f'(h(t)) = h'(t+1)/h'(t) \rightarrow e$ as $t \rightarrow -\infty$. Thus the extension is C^1 at zero and consequently C^1 on $\mathbb{R}$. It is nonlinear, because its minimal polynomial has degree three.

For every $\alpha > 0$, however,

$$\frac{|f(h(t)) - eh(t)|}{h(t)^{1+\alpha}} = \frac{e|2t+1|e^{-\alpha t}}{(t^2+4)^{1+\alpha}} \rightarrow \infty \quad (t \rightarrow -\infty). \quad (68)$$

So the positive-order remainder hypothesis of Theorem 9.2 genuinely strengthens bare differentiability. The analytic quartic of Section 3, on the other hand, shows that even real analyticity cannot restore a universal elimination rule at radius one, where the map need not have any fixed point.

10 Exact verification and reproducible artifacts

10.1 What the exact program checks

The script `code/verify.py` requires only Python 3.9 or later and its standard library. All algebraic checks use `Fractions.Fraction`; there is no floating-point arithmetic in these certificates. It implements the cubic function in two independent ways: the direct two-branch scaled formula and the polygonal conjugacy (14).

The recorded run tested 4,026 distinct rational starting points, including positive and negative points, multiple scales, and exact breakpoints. At every point it checked the cubic identity, oddness, scaling, agreement of the two implementations, and inverse identities. It also checked 4,025 adjacent exact secant slopes against the bounds $1/2$ and 4 . These finite tests supplement the all-real branch proof; they are not its logical replacement.

The program separately verifies recurrences on the integer range $-80 \leq n \leq 80$ and computes nonzero exact Hankel determinants for four sample orbits. The symbolic exponential-polynomial arguments explain why the recurrences hold for every integer and why the determinants certify the stated minimal orders.

Table 2: Exact sample certificates. The companion JSON file records the full matrices and polynomial coefficients.

Minimal polynomial	Degree	Hankel determinant
$z^3 - 8$	3	-56
$(z - 1)^2(z^2 + 1)$	4	1
$(z - 2)^3$	3	-512
$(z - 2)^5(z + 2)^4(z^2 + 4)^3$	15	nonzero exact integer

For the degree-15 example the sampled coordinate is

$$h(n) = 2^n \left((n^2 + 64)^2 + \frac{n^2}{1024} \cos \frac{\pi n}{2} + \frac{n^3}{1024} \cos \pi n \right). \quad (69)$$

Every sampled trigonometric value is an integer, so the whole sequence is rational. The real-variable version satisfies the derivative bounds in Section 6.4. Indeed, $k = 2$, $C = 64$, $1/2 < \log 2 < 1$, and $\pi < 4$ give

$$\frac{k}{\sqrt{C}} = \frac{1}{4} < \frac{\log 2}{2}, \quad \frac{2 \log 2 + 5 + 3\pi/2}{1024} < \frac{13}{1024} < \frac{\log 2}{4}. \quad (70)$$

Thus this is not merely a recurrence sequence detached from a monotone function; the construction realizes it as an actual orbit.

10.2 Numerical illustration, clearly separated

The optional script `code/analytic_demo.py` evaluates (17) by bisection. Since $|h(t) - t| \leq 1/4$, the inverse of x is bracketed by $x - 1/4$ and $x + 1/4$ before iteration begins. On 2,001 points in $[-10, 10]$, the recorded double-precision run gave a maximum absolute quartic residual of approximately $7.11 \cdot 10^{-15}$ and a maximum absolute residual in $f^{\circ 4}(x) = x + 4$ of approximately $1.24 \cdot 10^{-14}$. These numbers are implementation checks, not error bounds valid for all inputs and not part of the proof.

The plotting option generates the two vector figures used in this paper. The exact script does not require a plotting library. The article's counterexample remains independently checkable using the three-row itinerary table and the small integer or rational determinants.

11 Conclusions and the remaining boundary

The selected root-elimination question has a definite negative answer under its stated general hypotheses. The self-similar cubic has a real root and a nonreal pair at the same maximal modulus, yet every one of those roots is required by a minimal relation. The analytic quartic shows that this conclusion is not an artifact of a nonsmooth example.

The circular-spectrum classification identifies the mechanism more precisely. Away from radius one, positivity of the normalized two-sided orbit requires an even maximal polynomial degree, producing an odd multiplicity at the positive root. On the unit circle, it is the orbit *increment* that must have constant sign; taking the increment lowers the degree at one, producing an even multiplicity there and a strict multiplicity margin over every oscillating mode. Small trigonometric perturbations realize all and only the permitted patterns.

Several boundaries remain outside this report. We have not classified minimal polynomials with roots on several distinct circles. We have not classified decreasing homeomorphisms, for which consecutive iterates reverse order. Nor have we handled noninjective equations with zero constant coefficient or arbitrary proper interval domains. These are limitations of the proved statements, not qualifications needed by the two counterexamples on $\mathbb{R}$. [Added 29 September 2026, batch 41: Part II classifies minimal polynomials with roots on several distinct circles for increasing homeomorphisms of $\mathbb{R}$ (Theorem 12.2) and gives the exact universal reduction of every equation with nonzero constant coefficient over its increasing solutions (Theorem 16.3). Decreasing homeomorphisms, zero constant coefficient and proper interval domains remain outside both parts.] [Added 29 September 2026, batch 42: Part III classifies the minimal polynomials of decreasing homeomorphisms of $\mathbb{R}$ (Theorem 25.1) and gives the exact universal reduction over decreasing solutions (Theorem 30.1); it also shows that a decreasing homeomorphism of a bounded interval satisfying such an equation is an involution (Corollary 27.4). Zero constant coefficient remains open in both orientations, and proper interval domains remain open apart from that corollary.]

Finally, mathematical verification and historical priority are different questions. The paper gives complete arguments and exact finite certificates; it does not supply an independent referee report or an exhaustive priority search. The broader linearity problem is not declared solved. What is established here is the failure of the proposed general nonreal-root deletion and an exact, constructive classification in the increasing single-circle setting. [Added 29 September 2026, batch 41: Part II answers the increasing, nonzero-constant-coefficient version of the linearity problem for arbitrary root moduli (Theorem 17.1); the problem for decreasing maps is still not declared solved.] [Added 29 September 2026, batch 42: Part III answers the decreasing version (Theorem 30.4); together the two parts answer the linearity problem for every P with $P(0) \neq 0$ (Remark 30.5). Zero constant coefficient remains open.]

Part II. The increasing spectral core on several circles

Sections 1–11 and Appendices A–B form the original report (Part I) of 20 September 2026, unchanged except for a second subtitle line, a paragraph on the title page and dated pointers to this addition (and, since batch 42, a third subtitle line, a second title-page paragraph and dated pointers to Part III). Part II was added on 29 September 2026 in batch 41 of ProveIt’s incoming-report intake, from one manuscript written against the text of Part I printed above. It answers the first boundary that Part I names in Section 11, “We have not classified minimal polynomials with roots on several distinct circles”, for increasing homeomorphisms of $\mathbb{R}$: Theorem 12.2 classifies the minimal iterative polynomials of increasing homeomorphisms with arbitrary root moduli, and Theorem 16.3 gives the exact universal order reduction for every prescribed equation. The other boundaries of that paragraph (decreasing maps, zero constant coefficient, proper intervals) remain open. [Added 29 September 2026, batch 42: decreasing maps of $\mathbb{R}$ with $P(0) \neq 0$ are treated in Part III.] Section 12 records the provenance and fixes the notation of the addition.

12 Provenance, conventions and the main results

12.1 One manuscript, one addition

Part II is batch-41 manuscript 02, *The Complete Increasing Spectral Core: Arbitrary-radius classification, optimal order reduction, and fixed-point spectra for polynomial iterative equations*, an AI-assisted research draft prepared for Vladimir Reshetnikov and dated 28 September 2026 (20 pages as delivered). It arrived as the archive `ProveIt_Increasing_Spectral_Core.zip` (inner directory `Increasing_Spectral_Core/`) in ProveIt commit 9754e8360 and was placed in eaf787d50. The manuscript is pinned not by a commit but by the Git blob of the file it inspected: Part I’s `article.tex` with blob ae6ea0aa0566316ba9ed06b1750c6e73275059da, retrieved through the GitHub content API on 28 September 2026. That blob is the one committed in dc54c3cb3 and was unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code, recorded outputs, status note and source audit are shipped beside Part I’s files with the prefix `02-spectral-core-` (Part I’s unprefixed files count as 01); its manuscript, PDF, delivery README and checksum ledger are not shipped and survive in the arrival commit.

The manuscript contributes, in the order printed here: the difference-spectrum identity (Section 13); a complete spectral test for positive bilateral recurrences, with an analytic realization (Section 14); the classification of minimal iterative polynomials of increasing homeomorphisms with arbitrary root moduli (Theorem 12.2, proved in Section 15); the explicit universal annihilator $\mathcal{R}(P)$ (Section 16); an exact affine and homogeneous-linear rigidity criterion (Section 17); criteria for realizations with and without fixed points (Section 18); explicit and uniformly bi-Lipschitz examples (Section 19); its finite checks and their limits (Section 20); ten research questions (Section 21); its conclusion (Section 22); a Hankel-certificate lemma (Section 23); and its source audit (Section 24). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- (a) The manuscript reproves, so as not to assume Part I, the lemmas on homeomorphisms, the annihilator

ideal, two-sided recurrence decomposition, orbitwise least common multiples, simultaneous returns and positive means, and the translation argument for $(z - 1)^2$. Its proofs are the same arguments as Part I's; they are printed once, in Part I, and credited where Part II uses them (Section 13.1, Lemma 14.1, Theorem 17.1). The one-circle paragraph of Proposition 14.2 repeats the parity argument of Theorem 5.3; it is kept, because the proposition covers several circles at once.

- (b) Remarks 15.3 and 17.2 were added in the merge. They show that on one circle Theorem 12.2 is Theorem 6.1 and Theorem 17.1 gives Corollaries 8.1 and 8.2; Remark 15.3 also records a finite comparison of the two classifications made in the write phase.
- (c) Four symbols were renamed to avoid collisions (Section 12.2); no normalization changed. The manuscript's references to "the ProveIt report" now read "Part I", and its file names are replaced by the shipped names.
- (d) The manuscript's appendices (a Hankel lemma and its source audit) are printed as the last two sections of Part II, so that Part I's appendices keep their letters.

Status (manuscript 02). The classification is for continuous increasing maps $\mathbb{R} \rightarrow \mathbb{R}$ and polynomials with nonzero constant coefficient. The proofs are self-contained. This is an AI-assisted, unrefereed research draft; it is not Lean- or Rocq-verified, and no declaration in ProveIt formalizes any statement of Part I or Part II. The extension beyond Part I is explicit, but an exhaustive claim of historical priority is not made. Part II does not classify decreasing solutions, arbitrary proper-interval domains, noninjective equations with zero constant coefficient, or all individual maps realizing an admissible spectrum; its code accepts factored spectra only and does not factor arbitrary polynomials; and no finite test in it certifies the infinite or analytic parts of the proofs.

12.2 Notation across the two parts

Part I's symbols keep their meanings in Part I. Part II introduces the difference spectrum, positive admissibility and two cores, and reuses several letters that Part I employs differently. Table 3 fixes the meanings used from here on; letters not listed (P , Q , μ_f , E , Ann , r) agree with Part I.

12.3 A specific continuation, not another circular example

Part I proves a circular-spectrum classification and counterexamples to unrestricted deletion of nonreal characteristic roots. Its conclusion expressly leaves spectra on several distinct circles unclassified. Its circular theorem distinguishes a nonunit circle from the unit circle, and its rigidity statements are correspondingly restricted. Part II addresses that stated gap.

The external background is the order-reduction program of Draga and Morawiec [1], including their Problem 6.1 about nonreal roots tied in modulus with a real root. Draga's subsequent note [2] treats additional elimination cases. Petrov [3] asks for polynomial conditions forcing continuous solutions to be linear. We answer the *increasing, nonzero-constant-coefficient* version of the associated rigidity problem, distinguishing homogeneous linearity from affineness. We do not classify decreasing solutions or every individual solution.

No theorem of Part I is assumed without proof. The classical orbit-recurrence framework and the circular positivity mechanism are reproved in the manuscript, because the passage from them to an arbitrary-radius theorem requires a new sufficiency construction and careful treatment of the root 1; in this report those reproofs are Part I's arguments and are cited there.

Table 3: Symbols of Part II, with the tempting false reading beside the true one.

Symbol	Meaning in Part II	Not to be confused with
$\mathcal{D}A$	A with <i>one</i> copy of $z - 1$ removed, if present ((72)); the <i>difference spectrum</i>	the dilations D_2, D_8 of Section 2; removal of the whole $(z - 1)$ -primary factor
$\delta(A)$	1 if $A(1) = 0$, else 0	—
$m_A(\lambda)$	multiplicity of the root λ in A	the same notion as Part I's m_λ, m_r, m_1
positive-admissible	Definition 12.1	“realizable”: μ is a minimal polynomial iff $\mu = z - 1$ or $\mathcal{D}\mu$ is positive-admissible (Theorem 12.2)
r_-, r_+	least and greatest root moduli of the polynomial in question (usually $\mathcal{D}\mu$)	Part I's single common radius r
$\mathcal{R}(P)$	the increasing spectral core (Definition 16.1)	the real line $\mathbb{R}$
$\mathcal{K}(B)$	the positive-recurrence core (Section 16)	—
κ	the largest odd integer $\leq m_B(r)$ in (85) (<i>renamed</i> from the manuscript's k)	Part I's k , where $m_r = 2k + 1$
$b_- < b_+$	least and greatest positive roots of B in (86) (<i>renamed</i> from $a < b$)	the logarithmic rates $a < b$ of Lemma 14.3
Λ	the large constant of Lemma 14.3 (<i>renamed</i> from T)	a trigonometric sequence $T(n)$; the translation $T(t) = t + 1$
Φ	the glued fixed-point map in Theorem 18.4 (<i>renamed</i> from G)	the envelope G of Lemma 14.3
A	a monic polynomial (a spectrum), usually $\mathcal{D}\mu$	Part I's polynomial <i>function</i> $A(t)$ in (40), (49)
G, W	two-ended envelope and weight, (76), (77); $W = 2^t + 2^{-t}$ in Theorem 19.1	Part I's perturbed envelope $W(t)$ in (40), (53)
g, h	positive coordinate $g = h'$ and its antiderivative	Part I's h plays the same role (a conjugating coordinate)
F	the conjugated map $F(h(t)) = h(t + 1)$ on $h(\mathbb{R})$; the polygonal map of Section 19	Part I's cubic counterexample F (Theorem 2.1)
H_λ, H	real modes (79) and their sum; H is also a polygonal interpolation in Section 19	Part I's conjugating map H of (14)
$\mu_<, \mu_>$	monic factors of μ from roots of modulus < 1 , > 1	—
$\text{Fix}(f)$	the fixed-point set of f	—

12.4 Conventions

Write $f^{\circ 0} = \text{id}$ and, for a real polynomial P ,

$$P[f](x) = \sum_{j=0}^{\deg P} a_j f^{\circ j}(x), \quad P(z) = \sum_{j=0}^{\deg P} a_j z^j. \quad (71)$$

An increasing map is strictly increasing. A map is globally bi-Lipschitz if there are $0 < c \leq C < \infty$ such that $c|x - y| \leq |f(x) - f(y)| \leq C|x - y|$ for all x, y . When a nonzero polynomial annihilates f , its monic least-degree annihilator is denoted by μ_f .

For a nonzero monic polynomial A with $A(0) \neq 0$, set

$$\delta(A) = \begin{cases} 1, & A(1) = 0, \\ 0, & A(1) \neq 0, \end{cases} \quad \mathcal{D}A = \frac{A}{(z-1)^{\delta(A)}}. \quad (72)$$

Thus $\mathcal{D}$ removes *one copy*, not the entire 1-primary factor. For a root λ , write $m_A(\lambda)$ for its multiplicity in A .

Definition 12.1 (Positive-admissible spectrum). A nonconstant monic real polynomial A with $A(0) \neq 0$ is *positive-admissible* if the following conditions hold. Let r_- and r_+ be the least and greatest moduli of its roots. At each extreme radius $r \in \{r_-, r_+\}$, the positive real number r is a root, and

$$m_A(r) = \max_{|\lambda|=r} m_A(\lambda). \quad (73)$$

If $r_- = r_+$, this multiplicity must also be odd. There is no additional condition at a strictly intermediate radius. The constant polynomial 1 is not positive-admissible.

Theorem 12.2 (Complete increasing-spectrum classification). *Let μ be nonconstant, monic, and real, with $\mu(0) \neq 0$. The following are equivalent:*

- (i) $\mu = \mu_f$ for an increasing homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$.
- (ii) Either $\mu = z - 1$, or $\mathcal{D}\mu$ is positive-admissible.
- (iii) $\mu = \mu_f$ for a globally bi-Lipschitz increasing homeomorphism, real analytic outside at most one point.

When $\mu \neq z - 1$ and the extreme root moduli of $\mathcal{D}\mu$ bracket 1, the realization in (iii) is real analytic everywhere and has no fixed point.

The phrase “at each extreme radius” is essential. When two or more circles occur, even multiplicities at both extremes are allowed. For example, $(z - 2)^2(z - 4)^2$ is realizable, although $(z - 2)^2$ alone is not. The root 1 must be processed before deciding how many circles remain: $(z - 1)(z - 2)^2$ is not realizable as a minimal polynomial.

12.5 Optimal universal reduction

For any real P with $P(0) \neq 0$, we will construct a monic divisor $\mathcal{R}(P)$ with the exact universal property

$$[P[f] = 0 \implies Q[f] = 0 \text{ for every continuous increasing } f : \mathbb{R} \rightarrow \mathbb{R}] \iff \mathcal{R}(P) \mid Q. \quad (74)$$

If increasing solutions exist, a single globally bi-Lipschitz map with minimal polynomial $\mathcal{R}(P)$ witnesses the sharpness of every retained root and multiplicity simultaneously. Section 16 gives the explicit formula; no search through all polynomial divisors is necessary.

12.6 What is added and what is inherited

Question	Part I	Part II
Minimal spectra	One modulus circle	Arbitrary finite sets of circles
Multiplicity restrictions	Circular parity and dominance	Extreme-circle rules and unrestricted interior multiplicities
Universal reduction	Circular rigidity tests	Exact smallest common annihilator for every P
Realizations	Circular coordinates	Two-ended analytic coordinates and global bi-Lipschitz extension
Fixed points	Selected regularity results	Separate spectral existence criteria with and without fixed points

13 From a nonlinear map to a finite-dimensional recurrence

13.1 What is taken from Part I

The manuscript opens by reproving four facts that Part I proves by the same arguments; they are used here in Part I's form.

- (1) A continuous solution $f : \mathbb{R} \rightarrow \mathbb{R}$ of $P[f] = 0$ with $P(0) \neq 0$ is a strictly monotone homeomorphism (Lemma 4.3).
- (2) $\text{Ann}(f)$ is an ideal of $\mathbb{R}[z]$, generated by μ_f when it is nonzero, because $(zP)[f](x) = P[f](f(x))$ uses *right composition* (Lemma 4.2, (25)). One must not distribute f over a linear combination of its values.
- (3) Every bilateral sequence annihilated by $A(E)$, $A(0) \neq 0$, has a unique exponential-polynomial expression $u(n) = \sum_{\lambda} p_{\lambda}(n)\lambda^n$ with $\deg p_{\lambda} < m_A(\lambda)$, and its minimal recurrence polynomial is $\prod_{p_{\lambda} \neq 0} (z - \lambda)^{1 + \deg p_{\lambda}}$ (Lemma 4.4).
- (4) μ_f is the least common multiple of the minimal recurrence polynomials of the orbits $u_x(n) = f^{\circ n}(x)$, $n \in \mathbb{Z}$, and finitely many orbits attain every multiplicity (Lemma 4.5).

The manuscript adds two observations. For a homeomorphism, a factor z could be cancelled from an annihilator by surjectivity; consequently $\mu_f(0) \neq 0$. And if E acts on functions by $E\nu = \nu \circ f$, then on the finite-dimensional cyclic space generated by id it is an invertible linear operator whose cyclic-vector minimal polynomial is μ_f (compare the remark after Lemma 4.5); the same letter E denotes the shift $n \mapsto n + 1$ on sequences, because evaluation intertwines the two actions. The nonlinear nature of f as a real map does not interfere with this linear action on functions.

13.2 Why exactly one factor at one disappears

Lemma 13.1 (Difference-spectrum identity). *For a homeomorphism with minimal polynomial μ , the function $f - \text{id}$ has minimal polynomial $\mathcal{D}\mu$ under right composition by f . Equivalently, $\mathcal{D}\mu$ is the least common multiple of the minimal recurrence polynomials of*

$$d_x(n) = f^{\circ n+1}(x) - f^{\circ n}(x). \quad (75)$$

If $f = \text{id}$, the difference function is zero and its monic annihilator is the constant polynomial 1.

Proof. The difference function is $(E - 1) \text{id}$. A polynomial Q annihilates it if and only if $\mu \mid (z - 1)Q$. The least monic such Q is $\mu / \gcd(\mu, z - 1)$, which is precisely $\mathcal{D}\mu$. Evaluation at every point gives the scalar formulation. $\square$

For increasing f , each nonzero sequence d_x has one strict sign for all $n \in \mathbb{Z}$. Apply $f^{\circ n}$ to the strict inequality $f(x) > x$ or $f(x) < x$. This sign property is the bridge between dynamics and positive recurrence spectra. (Part I uses the same increments (36) on the unit circle only.)

14 Positive bilateral recurrences: a complete spectral test

14.1 The elementary oscillation argument

A finite trigonometric sequence is a finite sum $T(n) = \sum_{\zeta \in S} c_\zeta \zeta^n$ with distinct $|\zeta| = 1$. Two elementary facts will be used.

Lemma 14.1. *There exist integers $N_k \rightarrow +\infty$ such that $\zeta^{N_k} \rightarrow 1$ for every $\zeta \in S$. If a real trigonometric sequence satisfies $\liminf_{n \rightarrow +\infty} T(n) \geq 0$, then $T(n) \geq 0$ for all $n \in \mathbb{Z}$. If such a sequence is nonzero and nonnegative, its constant-frequency coefficient is strictly positive.*

Proof. The first assertion, and the returns $T(N_k + \ell) \rightarrow T(\ell)$ and $T(-N_k + \ell) \rightarrow T(\ell)$ in both time directions, are Lemma 5.1; the second assertion follows from them exactly as in the proof of Theorem 5.3; the third is Lemma 5.2. The manuscript proves all three by the same compactness and Cesàro-mean arguments. $\square$

Proposition 14.2 (Necessary extreme-circle conditions). *The minimal polynomial of a nonzero nonnegative real bilateral recurrence sequence is positive-admissible.*

Proof. Let r_+ be the greatest active root modulus and d_+ the greatest polynomial degree on that circle. As $n \rightarrow +\infty$,

$$u(n) = r_+^n n^{d_+} (T_+(n) + o(1)),$$

where T_+ is the nonzero trigonometric sequence made from the leading coefficients on that circle. Positivity and Lemma 14.1 show that $T_+ \geq 0$ and its constant frequency is positive. Thus the root r_+ occurs with degree d_+ , giving maximal multiplicity.

At $n \rightarrow -\infty$, the least active modulus r_- dominates instead. Apply the same argument to $u(-m)$ as $m \rightarrow +\infty$. The positive root r_- must again attain the greatest multiplicity on its circle.

If there is only one circle, let d be its maximum degree and T its leading trigonometric sequence. Positive time gives $T(\ell) \geq 0$; negative time, using the same simultaneous returns, gives $(-1)^d T(\ell) \geq 0$ for all integers ℓ . Since $T \neq 0$, d is even. The positive-root multiplicity $d + 1$ is therefore odd. (This paragraph is the argument of Theorem 5.3.) $\square$

The two distinct extreme circles use different leading coefficients, so there is no parity relation between them. A parity condition imposed separately on every extreme circle would be false.

14.2 A positive analytic interpolation with any allowed spectrum

Lemma 14.3 (Two-ended positive envelope). *Let $a < b$ and let d, e be arbitrary nonnegative integers. For sufficiently large $\Lambda > 0$, the function*

$$G(t) = e^{at}(\Lambda - t)^d + e^{bt}(\Lambda + t)^e \quad (76)$$

is strictly positive on $\mathbb{R}$ and satisfies

$$cW(t) \leq G(t) \leq CW(t), \quad W(t) = e^{at}(1 + |t|)^d + e^{bt}(1 + |t|)^e, \quad (77)$$

for some constants $c, C > 0$.

Proof. On $[-\Lambda, \Lambda]$ both summands are nonnegative and they do not vanish simultaneously. For $t \geq \Lambda$, the second summand is positive. The first can be negative only when d is odd, and then the ratio of its absolute value to the second is bounded by

$$e^{-(b-a)t} t^{d-e}.$$

Choose $\Lambda \geq 1$ large enough that this bound is at most $1/2$ for all $t \geq \Lambda$. At $t \leq -\Lambda$, the first summand is positive, and a potentially negative second summand is controlled by $e^{-(b-a)|t|} |t|^{e-d}$. Enlarge Λ to make that bound at most $1/2$. This proves positivity everywhere. At $-\infty$, $G/W \rightarrow 1$; the same is true at $+\infty$. A strictly positive continuous quotient has positive minimum and finite maximum on the remaining compact interval, giving (77). $\square$

Theorem 14.4 (Positive recurrence realization). *A nonconstant monic real polynomial A with $A(0) \neq 0$ is positive-admissible if and only if it is the minimal polynomial of a strictly positive bilateral recurrence sequence. Moreover, every such A is the minimal shift polynomial of a strictly positive real-analytic exponential polynomial $g : \mathbb{R} \rightarrow (0, \infty)$ such that*

$$0 < c \leq \frac{g(t+1)}{g(t)} \leq C < \infty \quad (t \in \mathbb{R}). \quad (78)$$

Here a shift polynomial annihilates a function of t under $t \mapsto t + 1$.

Proof. Necessity is Proposition 14.2. For sufficiency, first suppose there are at least two root moduli. Put $a = \log r_-$, $b = \log r_+$, $d = m_A(r_-) - 1$, and $e = m_A(r_+) - 1$. Use G from Lemma 14.3 as the two positive-real-root parts.

For every remaining real root or conjugate pair, choose the representative $\lambda = \rho e^{i\theta}$ with $0 \leq \theta \leq \pi$ and insert the real term

$$H_\lambda(t) = e^{(\log \rho)t} t^{m_A(\lambda)-1} \cos(\theta t). \quad (79)$$

For positive real roots $\theta = 0$; for negative ones $\theta = \pi$. On an extreme circle, the degree condition bounds $|H_\lambda|/W$. For a strictly interior radius, exponential separation at both ends bounds the same quotient, regardless of polynomial degree. Thus, with $H = \sum H_\lambda$, there is $M < \infty$ such that $|H| \leq MW$. Choose a nonzero positive ε small enough that $g = G + \varepsilon H \geq cW/2$. All intended leading mode coefficients are nonzero, and no new mode has been introduced.

If there is only one radius r , write $a = \log r$ and $d = m_A(r) - 1$. The admissibility condition makes d even. Start instead with

$$G(t) = e^{at}(1 + t^2)^{d/2}.$$

Each remaining term (79) has polynomial degree at most d , so a sufficiently small nonzero perturbation again stays between two positive multiples of G .

To justify exact minimality under a real shift, write cosine terms as pairs of complex exponentials. On $e^{st} p(t)$, $E - e^s$ lowers polynomial degree by one. Distinct continuous exponents give independent functions. The exponents $s = \log \rho \pm i\theta$ have $e^s = \lambda, \bar{\lambda}$; when $\theta = \pi$ both give the same negative root, but their leading coefficient functions cannot cancel identically. The maximal degree in that primary component is still $m_A(\lambda) - 1$. It follows that the minimal shift polynomial is exactly A .

The weights $W(t+1)/W(t)$ are bounded above and away from zero: $1 + |t+1|$ and $1 + |t|$ differ by factors between $1/2$ and 2 , and the exponential factors acquire fixed multipliers. The same assertion holds for the single-circle weight. Comparability gives (78).

Finally, for each root the leading coefficient of the sampled sequence $g(t_0 + n)$ is a nonzero real-analytic function of t_0 (or a nonzero complex-valued analytic function). Avoid the finitely many zero sets. Then $g(t_0 + n) > 0$ has minimal polynomial A . $\square$

The last generic-sampling step matters for negative roots. An antiderivative can contain $\sin(\pi t)$, which vanishes at every integer although it is not the zero function. Exact minimality must be proved as a function of the continuous coordinate, or at a nonsingular sampling phase, not inferred carelessly from phase zero.

15 Proof of the increasing-spectrum classification

15.1 Necessity for a family of orbit increments

Lemma 15.1. *The least common multiple of finitely many positive-admissible polynomials is positive-admissible.*

Proof. Take any root at an extreme circle of the least common multiple, with its attained maximal multiplicity. In a factor attaining it, this circle is also extreme; that factor supplies the positive root with at least that multiplicity. If the union has only one circle, every factor has that circle, and the maximum of its positive-root multiplicities is the maximum of finitely many odd integers, hence is odd. $\square$

Suppose $f \neq \text{id}$ is increasing with minimal polynomial μ . By Lemma 13.1, $\mathcal{D}\mu$ is the least common multiple of the nonzero increment-orbit minimal polynomials, using finitely many orbits. Multiply each increment sequence by its constant sign to make it positive. Proposition 14.2 and Lemma 15.1 show that $\mathcal{D}\mu$ is positive-admissible. For $f = \text{id}$ the minimal polynomial is $z - 1$. This proves the necessity in Theorem 12.2.

15.2 Integrating the positive coordinate

Assume $A = \mathcal{D}\mu$ is positive-admissible. Construct $g > 0$ as in Theorem 14.4. We choose a real-analytic antiderivative h whose minimal shift polynomial is exactly μ .

For a nonzero continuous exponent s , a canonical antiderivative is

$$\int e^{st} p(t) dt = e^{st} \sum_{j=0}^{\deg p} \frac{(-1)^j p^{(j)}(t)}{s^{j+1}}. \quad (80)$$

It preserves polynomial degree and hence the shift-root multiplicity. For the frequency $s = 0$, ordinary integration increases polynomial degree by one. Apply this term by term, respecting complex conjugation. If 1 is a root of A , the multiplicity at 1 increases by one, as required. If 1 is absent from A but present in μ , add a nonzero constant, say 1. If 1 is absent from μ , add no constant. Thus $h' = g > 0$ and its exact shift polynomial is μ .

Let $(\alpha, \beta) = h(\mathbb{R})$, with infinite endpoints permitted. Define on this interval

$$F(h(t)) = h(t + 1). \quad (81)$$

The analytic inverse function theorem makes F real analytic there, and

$$F'(h(t)) = \frac{g(t + 1)}{g(t)}. \quad (82)$$

Consequently it is bi-Lipschitz with constants uniform on the interval. Also $P[F](h(t)) = P(E)h(t)$ for every polynomial P , proving exact minimality on that interval.

15.3 The range and extension through a finite endpoint

Write r_- and r_+ for the extreme root moduli of A . The envelope asymptotics give the following exhaustive possibilities:

$$h(\mathbb{R}) = \begin{cases} \mathbb{R}, & r_- \leq 1 \leq r_+, \\ (c, \infty), & 1 < r_-, \\ (-\infty, c), & r_+ < 1. \end{cases} \quad (83)$$

Indeed, at an endpoint with exponential rate zero, g has a nondecaying polynomial envelope and is not integrable. A positive rate is integrable only at $-\infty$, and a negative rate only at $+\infty$. The extreme rates are ordered, so both endpoints cannot be finite.

When the range is $\mathbb{R}$, take $f = F$. It has no fixed point because $h(t + 1) > h(t)$. In a half-line case, set $f(c) = c$ and extend by reflection:

$$f(2c - x) = 2c - F(x) \quad (84)$$

for x in the original half-line. The limiting value of F at c is c , since both $h(t)$ and $h(t + 1)$ tend there. This extension is an increasing homeomorphism. Its bi-Lipschitz bounds hold on each side and pass across c by addition. It is analytic away from c .

If $\mu(1) \neq 0$, our choice of primitive gives $c = 0$. If $\mu(1) = 0$, the constant contribution c is annihilated. Thus in both cases $\mu(1)c = 0$. Iterating (84) gives

$$\mu[f](2c - x) = 2c\mu(1) - \mu[F](x) = 0.$$

The value at c satisfies the same equation. Minimality on the original half-line forces minimality globally. This proves sufficiency and all regularity claims of Theorem 12.2.

Remark 15.2 (What is not being claimed about analyticity). The construction is globally real analytic when its coordinate covers $\mathbb{R}$. In a half-line case the reflected map need not be differentiable at c . No complex-entire inverse is used, and no global complex analyticity is asserted. These distinctions avoid an otherwise incorrect regularity upgrade at an accumulating fixed point.

Remark 15.3 (Added in the merge: one circle recovers Part I). When all roots of μ lie on one circle $|z| = r$, Theorem 12.2 is Theorem 6.1. If $r \neq 1$, then $\mathcal{D}\mu = \mu$ has one circle, and positive admissibility says that r is a root of odd multiplicity at least every other multiplicity: condition (i) of Theorem 6.1. If $r = 1$ and $\mu \neq z - 1$, then $\mathcal{D}\mu$ is nonconstant and lies on the unit circle; it is positive-admissible exactly when 1 is a root of $\mathcal{D}\mu$ of odd multiplicity $m_\mu(1) - 1 \geq 1$ that is at least every other multiplicity, that is, when $m_\mu(1)$ is even, at least 2, and exceeds every other multiplicity: condition (ii). The realizations agree in kind as well: by Theorem 18.3 below, for $\mu \neq z - 1$ on one circle a fixed-point-free realization exists exactly when $r = 1$, in accordance with Section 6 (every realization fixes 0 when $r \neq 1$ and has no fixed point when $r = 1$). As a finite check of the implementation, the manuscript's `realizable` test (shipped as `code/02-spectral-core-spectral_core.py`) was compared in the write phase with the conditions of Theorem 6.1 on 638 single-circle spectra: radii 2 and 1; the roots $\pm r$, the pair $\pm ri$ and the pair $r(3 \pm 4i)/5$; the positive root with multiplicity 0 to 4 and the other three with multiplicity 0 to 3, the empty spectrum excluded (2×319 cases). There were no disagreements. The comparison script is not shipped; the grid above specifies it completely.

16 The exact universal spectral core

16.1 A root-by-root formula

Normalize P to be monic, which does not change its solution set. First define a positive-recurrence core $\mathcal{K}(B)$ for any monic real B with $B(0) \neq 0$.

If B has no positive real root, set $\mathcal{K}(B) = 1$. If its only distinct positive root is r , put

$$\kappa = 2 \left\lfloor \frac{m_B(r) - 1}{2} \right\rfloor + 1, \quad \mathcal{K}(B) = \prod_{|\lambda|=r} (z - \lambda)^{\min(m_B(\lambda), \kappa)}. \quad (85)$$

Thus κ is the largest odd integer not exceeding $m_B(r)$.

If there are at least two distinct positive roots, let $b_- < b_+$ be the least and greatest such roots. Define

$$k_B(\lambda) = \begin{cases} 0, & |\lambda| < b_- \text{ or } |\lambda| > b_+, \\ \min(m_B(\lambda), m_B(b_-)), & |\lambda| = b_-, \\ m_B(\lambda), & b_- < |\lambda| < b_+, \\ \min(m_B(\lambda), m_B(b_+)), & |\lambda| = b_+, \end{cases} \quad (86)$$

and set $\mathcal{K}(B) = \prod_{\lambda} (z - \lambda)^{k_B(\lambda)}$. Conjugate roots receive identical multiplicities, so these are real polynomials.

Definition 16.1 (Increasing spectral core). For monic P with $P(0) \neq 0$, define

$$\mathcal{R}(P) = (z - 1)^{\delta(P)} \mathcal{K}(\mathcal{D}P). \quad (87)$$

The value $\mathcal{R}(P) = 1$ means that no increasing solution exists.

Lemma 16.2. *Every positive-admissible divisor of B divides $\mathcal{K}(B)$. When $\mathcal{K}(B) \neq 1$, it is itself positive-admissible.*

Proof. For an admissible divisor, its extreme root moduli are positive roots of B . Thus all its roots lie between the least and greatest positive roots of B . At a boundary of that full interval, its multiplicities cannot exceed

the positive-root multiplicity there. This gives all bounds in (86). If B has just one positive root, the divisor has only that radius, and its positive-root multiplicity is odd. Formula (85) is exactly the largest permitted cap. Conversely the displayed cores satisfy the defining extreme and parity conditions directly. $\square$

Theorem 16.3 (Sharp universal order reduction). *For every real P with $P(0) \neq 0$, the polynomial $\mathcal{R}(P)$ satisfies:*

- (i) $\mathcal{R}(P) \mid P$, and for continuous increasing maps on $\mathbb{R}$, $P[f] = 0$ if and only if $\mathcal{R}(P)[f] = 0$.
- (ii) Increasing solutions exist if and only if $\mathcal{R}(P) \neq 1$; equivalently, if and only if P has a positive real root.
- (iii) When solutions exist, there is a globally bi-Lipschitz increasing map with minimal polynomial exactly $\mathcal{R}(P)$.
- (iv) For every real polynomial Q , the universal implication (74) holds exactly when $\mathcal{R}(P) \mid Q$.

Proof. Divisibility in (i) follows from the construction. Suppose $P[f] = 0$ and let $\mu = \mu_f$, so $\mu \mid P$. The operation $\mathcal{D}$ preserves divisibility: at the root 1 it replaces multiplicity m by $\max(m - 1, 0)$, a monotone function of m . If $f \neq \text{id}$, the classification makes $\mathcal{D}\mu$ admissible. Lemma 16.2 gives $\mathcal{D}\mu \mid \mathcal{K}(\mathcal{D}P)$. Restoring the possible one copy of $z - 1$ yields $\mu \mid \mathcal{R}(P)$, hence $\mathcal{R}(P)[f] = 0$. For $f = \text{id}$, the factor $z - 1$ in P and in $\mathcal{R}(P)$ gives the same result. The reverse implication follows because $\mathcal{R}(P)$ divides P .

If $\mathcal{R}(P) \neq 1$ but $\mathcal{K}(\mathcal{D}P) = 1$, then $\mathcal{R}(P) = z - 1$, realized by the identity. Otherwise $\mathcal{D}\mathcal{R}(P) = \mathcal{K}(\mathcal{D}P)$: when P has no root at one, neither does its positive core. Theorem 12.2 gives (iii). If $\mathcal{R}(P) = 1$, (i) would force $\text{id} = 0$, impossible on $\mathbb{R}$. A positive root r supplies the solution $f(x) = rx$, and the formula has no nontrivial value without a positive root. This proves (ii).

If $\mathcal{R}(P) \mid Q$, every solution of the reduced equation satisfies $Q[f] = 0$. Conversely, apply any proposed universal relation to the map in (iii); its minimal polynomial forces $\mathcal{R}(P) \mid Q$. When there are no solutions, the implication is vacuous and 1 divides every Q , so (iv) remains valid. $\square$

Equivalently, the ideal of common iterative relations of the increasing solution class is

$$\bigcap_{\substack{f: P[f]=0 \\ f \text{ continuous increasing}}} \text{Ann}(f) = (\mathcal{R}(P)). \quad (88)$$

The intersection over an empty class is the full polynomial ring. This is stronger than merely exhibiting some eliminable roots: it identifies the exact obstruction to *every* further universal reduction.

16.2 Consequences for boundary and interior roots

When $\mathcal{D}P$ has two distinct positive roots, every root strictly between the smallest and largest positive radii survives at its full multiplicity in the universal core. At a boundary, a nonreal or negative root survives up to the multiplicity of the positive root there. It need not disappear. When only one positive root remains, the additional odd cap is exact. The unit root affects these conclusions only through the preliminary one-step multiplicity reduction, but omitting that step changes answers.

The core is idempotent:

$$\mathcal{R}(\mathcal{R}(P)) = \mathcal{R}(P). \quad (89)$$

It also respects divisibility: $P \mid Q$ implies $\mathcal{R}(P) \mid \mathcal{R}(Q)$. This follows either from the sharp universal property or by observing that realizable divisors of P are realizable divisors of Q .

17 An exact answer to increasing affine rigidity

The distinction between a polynomial being the minimal polynomial of *an affine map* and forcing *all maps* to be affine cannot be ignored. A small piecewise-linear example is decisive.

Theorem 17.1 (Affine and homogeneous-linear rigidity). *Assume that P has at least one increasing continuous solution on $\mathbb{R}$. Then all such solutions are affine if and only if*

$$\mathcal{R}(P) = z - r \text{ for some } r > 0, \quad \text{or} \quad \mathcal{R}(P) = (z - 1)^2. \quad (90)$$

All such solutions are homogeneous linear if and only if $\mathcal{R}(P) = z - r$ for some $r > 0$. In that case the unique increasing solution is $f(x) = rx$. When $\mathcal{R}(P) = (z - 1)^2$, the solutions are exactly the translations $f(x) = x + c$.

Proof. A reduced polynomial $z - r$ forces $f = r \text{ id}$. If $(z - 1)^2[f] = 0$, the argument in the proof of Corollary 8.2 (each bilateral orbit is $f^{on}(x) = x + n(f(x) - x)$, and order preservation for every $n \in \mathbb{Z}$ forces the coefficient of n in $(y - x) + n((f(y) - y) - (f(x) - x)) > 0$ to vanish) shows that $f(x) - x$ is constant, and every translation does satisfy the equation. The manuscript gives the same argument.

For necessity, let $R = \mathcal{R}(P)$ and use a realization with $\mu_f = R$. An increasing affine map $ax + b$ has minimal polynomial $z - a$ if $b = 0$, $(z - 1)^2$ if $a = 1, b \neq 0$, and $(z - 1)(z - a)$ if $a \neq 1, b \neq 0$. Thus unless R has one of these forms, the realization is nonaffine. The remaining case $R = (z - 1)(z - r)$ with $r > 0, r \neq 1$, also fails universal affine rigidity. Indeed

$$f(x) = \begin{cases} x, & x \leq 0, \\ rx, & x \geq 0 \end{cases} \quad (91)$$

is increasing and continuous, satisfies $R[f] = 0$, and is not affine. This completes the affine classification. Finally nonzero translations are not homogeneous linear, proving the last assertions. $\square$

The piecewise-linear mechanism is already visible in Bogdanov's comment on Petrov's question [3]; it is not claimed as a new construction. What the present result adds is an exact test for arbitrary input spectra, including multiplicities and nonreal roots, through $\mathcal{R}(P)$. The unrestricted version of Petrov's question includes decreasing maps and is not declared solved here. [Added 29 September 2026, batch 42: decreasing maps are decided in Part III, Theorem 30.4 and Remark 30.5; the version with $P(0) = 0$ remains open.]

Remark 17.2 (Added in the merge: the circular rigidity criteria). For P supported on one circle, Theorem 17.1 gives Corollaries 8.1 and 8.2. If $r \neq 1$ and r is a root, then $\mathcal{R}(P) = \mathcal{K}(P)$ is given by (85), and it equals $z - r$ exactly when $\kappa = 1$ and P has no other root, that is, $P = (z - r)^m$ with $m \in \{1, 2\}$. If $r = 1$ and $M = m_P(1) \geq 1$, then $\mathcal{R}(P) = z - 1$ exactly when $\mathcal{D}P$ has no positive root, that is, $M = 1$; and $\mathcal{R}(P) = (z - 1)^2$ exactly when $\mathcal{K}(\mathcal{D}P) = z - 1$, that is, $P = (z - 1)^M$ with $M - 1 \in \{1, 2\}$. By Theorem 16.3(ii), $M = 0$ (or r not a root) leaves no increasing solution. These are the statements of the two corollaries.

18 Which spectra admit fixed points?

18.1 Finite limits eliminate incompatible modes

Lemma 18.1. *Suppose a real recurrence sequence $u(n)$ has a finite limit as $n \rightarrow +\infty$. Its active roots are either of modulus less than 1, or a simple root at 1. Its 1-component equals the limit. With a finite limit as $n \rightarrow -\infty$, replace “less” by “greater”. A recurrence sequence with finite limits at both ends is constant.*

Proof. A greatest-modulus, greatest-degree normalization, as in Proposition 14.2, shows that a nonzero component with modulus greater than one cannot remain bounded: a nonzero finite trigonometric leading term returns away from zero along a subsequence. Apply this successively to remove the finitely many growing levels. On the unit circle, positive polynomial degree is likewise impossible under boundedness. A degree-zero trigonometric sequence having a limit must be constant: its difference from that limit tends to zero, and the simultaneous returns in Lemma 14.1 make it identically zero. The negative-time claim follows by replacing n by $-n$. The intersection of the two permitted supports is the simple constant mode. $\square$

Proposition 18.2 (The fixed set is an interval). *For an increasing polynomial-iterative homeomorphism, $\text{Fix}(f)$ is empty or a closed interval, with points, rays, and the whole line allowed. In particular, a nontrivial bounded component of $\mathbb{R} \setminus \text{Fix}(f)$ cannot occur.*

Proof. The fixed set is closed. If $a < b$ are fixed, every orbit of a point $x \in [a, b]$ stays in $[a, b]$ for all integer times and is monotone. It has finite limits at both ends. Lemma 18.1 makes the orbit constant, so x is fixed. $\square$

18.2 Fixed-point-free realization

Theorem 18.3. *Let $\mu \neq z - 1$ satisfy Theorem 12.2, and let r_-, r_+ be the extreme root moduli of $\mathcal{D}\mu$. A realization with no fixed point exists if and only if*

$$r_- \leq 1 \leq r_+. \quad (92)$$

When it exists it can be globally real analytic and bi-Lipschitz.

Proof. Sufficiency is the full-line case of (83). For necessity, if all roots of $\mathcal{D}\mu$ have modulus greater than one, every increment sequence is absolutely summable toward $-\infty$. Thus every orbit has a finite backward limit, necessarily fixed by continuity. If all roots have modulus less than one, the forward limit is finite instead. Both contradict the absence of fixed points. $\square$

18.3 Fixed-point realization

For a polynomial μ , let $\mu_<$ and $\mu_>$ be its monic factors formed from roots with modulus respectively less and greater than one. An empty factor is 1.

Theorem 18.4 (Fixed-point spectra). *A nonconstant monic real μ , with $\mu(0) \neq 0$, is the exact minimal polynomial of an increasing homeomorphism having a fixed point if and only if either $\mu = z - 1$, or all the following hold:*

- (i) *Its only possible root on the unit circle is 1, with multiplicity at most one.*
- (ii) *At least one of $\mu_<, \mu_>$ is nonconstant.*
- (iii) *Each nonconstant member of $\{\mu_<, \mu_>\}$ is positive-admissible.*

These realizations can be chosen globally bi-Lipschitz and analytic away from one fixed point.

Proof. Suppose a nonidentity realization has a fixed point. By Proposition 18.2, each nonfixed component is a ray with a finite fixed endpoint c . Every orbit on it tends to c in one time direction: otherwise its finite limit would be another fixed point inside the ray. Lemma 18.1 shows that such an orbit has only roots strictly on one side of the unit circle, together with a possible simple constant mode. Consequently (i) holds globally. For increments, the constant mode disappears. The factors $\mu_<$ and $\mu_>$ are separately least common multiples of positive increment-orbit spectra, and hence are admissible by Lemma 15.1. Nonidentity gives (ii).

Conversely construct, for each nonconstant factor, a half-line map with that exact minimal polynomial and finite endpoint zero, using Section 15. Reflection or reversal of the coordinate moves its domain to either desired half-line. If both factors occur, put one on each side of zero. If only one occurs, use it on both sides. The combined map Φ fixes zero, is globally bi-Lipschitz, is analytic away from zero, and has minimal polynomial $\mu_<\mu_>$. There is no root at one in this product.

If μ includes a simple $z - 1$, translate the map: $f(x) = 1 + \Phi(x - 1)$. Any annihilator of f , evaluated at its fixed point 1, must vanish at one. Subtracting this constant contribution shows that it must also annihilate Φ . Conversely the product of those two coprime factors annihilates f . Its minimal polynomial is therefore $(z - 1)\mu_<\mu_>$, as required. $\square$

Theorem 18.4 is genuinely stronger than the general classification. For example $(z - \frac{1}{2})(z - 2)^2$ is admissible globally, but its exterior factor $(z - 2)^2$ is not. Every realization is therefore fixed-point-free. In contrast, $(z - \frac{1}{2})(z - 2)$ has both fixed-point and fixed-point-free realizations. A spectrum does not in general determine the dynamical type of every solution.

19 Explicit examples and arbitrarily high nonreal multiplicity

19.1 Even multiplicities on two extreme circles

Set, for every $n \in \mathbb{Z}$,

$$v_n = 2^n(2 - n) + 4^n(2 + n), \quad (93)$$

$$u_n = 2^n(4 - n) + 4^n\left(\frac{n}{3} + \frac{2}{9}\right). \quad (94)$$

Direct subtraction gives $u_{n+1} - u_n = v_n$. The value v_n is positive: for $|n| \leq 2$ both summands in (93) are nonnegative and not both zero; for $n > 2$ the positive $4^n(n + 2)$ exceeds $2^n(n - 2)$; and for $n < -2$ the corresponding inequality is reversed between the two bases. In fact the same argument works for all real n .

The sequence u_n increases from 0 to $+\infty$. Let H be its strictly increasing polygonal interpolation, $H(n + s) = (1 - s)u_n + su_{n+1}$ for $0 \leq s \leq 1$. Define $F(H(t)) = H(t + 1)$ on the positive half-line and extend oddly through zero. Each orbit at phase s is a linear combination of two shifted copies of u_n , so

$$((z - 2)^2(z - 4)^2)[F] = 0. \quad (95)$$

The exact determinant

$$\det(u_{i+j})_{0 \leq i, j \leq 3} = \frac{16384}{9} \neq 0 \quad (96)$$

certifies minimal degree four, hence the displayed minimal polynomial. The slopes on consecutive intervals are v_{n+1}/v_n . They have positive finite limits 2 and 4 at the two ends; positivity on the remaining finite set gives global bi-Lipschitz bounds. This rational polygonal example is supplemental to the analytic realization theorem, not a substitute for it.

19.2 An everywhere analytic annular example with rational orbit data

Let

$$h(t) = 2^t - 2^{-t} + \frac{1}{10} \sin \frac{\pi t}{2}, \quad f(h(t)) = h(t+1). \quad (97)$$

Since

$$h'(t) = (\log 2)(2^t + 2^{-t}) + \frac{\pi}{20} \cos \frac{\pi t}{2} \geq 2 \log 2 - \frac{\pi}{20} > 0,$$

the coordinate is an analytic bijection $\mathbb{R} \rightarrow \mathbb{R}$ with nonvanishing derivative. The last strict inequality follows, for example, from $\log 2 > 1/2$ and $\pi < 4$. The resulting f is analytic and fixed-point-free, and its minimal polynomial is

$$\mu_f(z) = (z - \frac{1}{2})(z - 2)(z^2 + 1) = z^4 - \frac{5}{2}z^3 + 2z^2 - \frac{5}{2}z + 1. \quad (98)$$

All terms of $h(n)$ are rational. The first four are 0, 8/5, 15/4, 311/40, and

$$\det(h(i+j))_{0 \leq i,j \leq 3} = \frac{225}{256} \neq 0. \quad (99)$$

Thus even the unit-circle nonreal pair survives without any root at one in the characteristic polynomial. The two real roots lie on opposite sides of the unit circle and supply the two ends of the coordinate. (Compare Part I's analytic quartic, Theorem 3.1, where the pair $\pm i$ needs the double root at one that Theorem 6.1 requires on a single circle.)

19.3 No multiplicity penalty inside the annulus

Theorem 19.1 (Uniformly bi-Lipschitz high-order nonreal modes). *For every integer $m \geq 1$, there is an everywhere real-analytic, fixed-point-free increasing homeomorphism f_m with*

$$\mu_{f_m}(z) = (z - \frac{1}{2})(z - 2)(z^2 + 1)^m, \quad \frac{1}{6} \leq f'_m(x) \leq 6 \quad (x \in \mathbb{R}). \quad (100)$$

In particular, a common global bi-Lipschitz bound places no upper bound on the multiplicity of an interior nonreal characteristic root.

Proof. Write $a = \log 2$, $k = m - 1$, and $\varepsilon = a^k / (2k!)$. Set

$$W(t) = 2^t + 2^{-t}, \quad g_m(t) = W(t) + \varepsilon t^k \cos \frac{\pi t}{2}. \quad (101)$$

The exponential-series inequality $e^{a|t|} \geq (a|t|)^k / k!$ implies

$$\varepsilon |t|^k \leq \frac{1}{2} e^{a|t|} \leq \frac{1}{2} W(t).$$

Hence $W/2 \leq g_m \leq 3W/2$. Take the canonical real antiderivative from (80), with no added constant. It is an increasing analytic bijection $h_m : \mathbb{R} \rightarrow \mathbb{R}$. Its exact shift polynomial is the one in (100): integration preserves degree k in the nonzero frequencies $\pm \pi i/2$ and introduces no zero frequency.

Define $f_m = h_m \circ (t \mapsto t+1) \circ h_m^{-1}$. Because $\frac{1}{2} \leq W(t+1)/W(t) \leq 2$,

$$\frac{1}{6} \leq \frac{g_m(t+1)}{g_m(t)} \leq 6.$$

Equation (82) proves the claimed derivative bounds. $\square$

The proof does not infer minimality from the orbit at $h_m(0)$. For certain parities an integrated oscillatory component can vanish at that sampling phase. The shift-space argument proves exact minimality for the map, and a generic phase supplies a full scalar-orbit certificate.

19.4 A complete order-32 to order-23 reduction

Consider the factored polynomial

$$P = (z - 1)^3(z - \frac{1}{2})^2(z - 2)^4(z + 2)^7(z^2 + 1)^5(z^2 + 9)^3. \quad (102)$$

After removing one copy at one, the least and greatest positive roots are 1/2 and 2, of multiplicities 2 and 4. The roots $\pm i$ are interior and keep multiplicity 5. The boundary root -2 is capped at 4, and $\pm 3i$ disappear. Restoring the root at one gives

$$\mathcal{R}(P) = (z - 1)^3(z - \frac{1}{2})^2(z - 2)^4(z + 2)^4(z^2 + 1)^5. \quad (103)$$

The degree falls from 32 to 23. Every increasing solution satisfies the reduced equation, and a single globally analytic bi-Lipschitz solution has exactly that degree-23 minimal polynomial. Therefore no further universal factor or multiplicity can be removed.

20 Reproducibility and verification boundaries

20.1 Exact code and the scope of its input

Part II ships a standard-library Python implementation, `code/02-spectral-core-spectral_core.py` (delivered as `code/spectral_core.py`). It accepts a factor-level spectrum of Gaussian-rational roots, with rational real and imaginary parts and integer multiplicities. Conjugation symmetry is validated. Squared moduli are rational, so every comparison is exact.

The implementation does *not* claim to factor an arbitrary input polynomial over the algebraic numbers. The theorem applies to all real coefficient polynomials; the shipped implementation has the narrower, explicit input representation above. For rational or real-algebraic coefficients, exact root isolation and comparisons of algebraic squared moduli would extend the front end without changing the core algorithm.

The manuscript's example below imports the module under its delivered name `spectral_core`. Under the shipped names that import fails; run it in a copy of `code/` with the delivered names restored, as the report's `README.md` describes, with that directory on Python's module search path (for example, `PYTHONPATH=code python`).

```
from fractions import Fraction
from spectral_core import Root, increasing_core

one = Root(1)
two = Root(2)
P = {one: 1, two: 2}
assert increasing_core(P) == {one: 1, two: 1}
```

The module also implements tests for exact realizability, fixed-point realizability, fixed-point-free realizability, and affine rigidity. It rejects zero roots, invalid multiplicities, and asymmetric conjugate spectra rather than silently applying the theorem outside its hypotheses.

20.2 Recorded exact checks

The verifier (`code/02-spectral-core-verify.py`, delivered as `code/verify.py`; the file of that name here is Part I's verifier) produced the following results, recorded in `data/02-spectral-core-exact_che`

cks.json and data/02-spectral-core-verification.txt.

Check	Count or exact result
Spectra in two finite multiplicity grids	1354
Divisors examined against the admissibility criterion	97281
Rational recurrence windows for each example	$-80 \leq n \leq 80$
Exact increment/monotonicity assertions	402
Hankel determinant for (94)	16384/9
Hankel determinant for (97)	225/256
Large-core degree reduction	$32 \rightarrow 23$

For each finite input spectrum, the verifier computes the core formula and independently enumerates all divisors satisfying the theorem’s admissibility test. It verifies equality with their least common multiple, core idempotence, divisibility, and exact realizability of every nontrivial core. This checks the finite combinatorics and implementation of the formula, not the analytic realization proof. (Merge note: the enumeration is independent of the core formula but uses the same `positive_admissible` routine as the rest of the module; the comparison with Part I’s independently stated circular theorem is the one in Remark 15.3.)

For each explicit orbit, rational arithmetic verifies the recurrence through the stated window and computes a nonzero Hankel determinant. The recurrence holds for all integers because of the symbolic exponential-polynomial formulas, not because of the finite window. The determinant certifies that no lower-order recurrence can annihilate that orbit; the all-real constructions connect those orbits to actual maps.

The worked example of Section 19 is also a script, `code/02-spectral-core-example.py` (delivered as `code/example.py`); its recorded output is `data/02-spectral-core-example.txt`. In the write phase (29 September 2026, Python 3.14.4) both scripts were rerun in a copy with the delivered names restored: both passed, and their outputs equal the shipped records apart from Windows line endings.

20.3 Rebuilding

The report’s `README.md` gives the rerun commands for both parts and the build command for this article. No bibliography processor is needed; Part II adds no figure files. The JSON output gives full exact matrices, polynomial coefficients in ascending order, and initial values. The short verification log is also included. The complete universal theorem is not machine-checked by these scripts, and no Lean certification is implied.

21 Further research questions and a formalization program

Q1. Decreasing spectra and failure of single-witness sharpness. Determine the exact minimal spectra and universal relation ideal for decreasing homeomorphisms. Simply replacing “positive root” by “negative root” is insufficient: odd-time order reversal and even-time monotonicity must hold simultaneously. Draga’s theorem [2] shows that certain two-negative-root equations reduce to a disjunction of lower-order equations. In such a class, the least common annihilator can require factors that are not simultaneously realized by any one map. The increasing case’s single-witness theorem may therefore have no direct decreasing counterpart. [Added 29 September 2026, batch 42: answered in Part III for continuous maps of $\mathbb{R}$ with $P(0) \neq 0$. The minimal spectra are those of Theorem 25.1, the universal relation ideal is generated by $C_{\downarrow}(P)$ (Theorem 30.1), and single-witness sharpness

does fail as predicted: exactly two witnesses are needed precisely when both a contracting and an expanding side core survive (Corollary 30.3), which includes Draga's disjunction. Proper intervals and $P(0) = 0$ remain open.]

- Q2. Smoothness at a fixed point with several radii.** Classify spectra realizable by globally C^k , $C^{k,\alpha}$, or real-analytic maps having a fixed point. The present construction is analytic off the point, and Part I (Section 9) establishes some single-circle regularity obstructions. Several radii introduce competing power scales and possible resonances. An exact threshold theorem would refine Theorem 18.4, rather than merely improve a construction. [Added 29 September 2026, batch 42: for *decreasing* maps, which always have a fixed point, Part III gives a finite threshold (Theorem 31.3) above which every non-involutive solution is analytic with the spectra of Theorem 32.1; the threshold is sufficient, not shown optimal. The increasing question stays open.]
- Q3. The least possible bi-Lipschitz distortion.** For an admissible μ , minimize a symmetric constant L in $L^{-1}|x-y| \leq |f(x)-f(y)| \leq L|x-y|$ over realizations with minimal polynomial μ . Extreme recurrence growth gives natural lower bounds, while Theorem 19.1 shows that large interior multiplicities need not force large distortion. Determine whether the infimum depends only on extreme radii, and when it is attained.
- Q4. All solutions, not only their possible spectra.** Describe the moduli of increasing solutions of a fixed equation. The realization theorem selects convenient coordinates but is not a parametrization of all solutions. One possible route is a positive cone of recurrence initial data together with an order-compatible choice of a fundamental interval and a gluing law.
- Q5. Convexity, concavity, and stronger order restrictions.** Find the analogue of $\mathcal{R}(P)$ for globally convex or concave increasing solutions. Convexity controls differences between nearby orbits, not just signs of individual increments. It may yield stronger multiplicity or oscillation restrictions than the positive-recurrence criterion.
- Q6. Proper intervals and one-sided iteration.** Classify self-maps of bounded intervals, half-lines, and proper open intervals without surjectivity. The proof here uses bilateral orbits in an essential way. A one-sided recurrence need not have the negative-time parity obstruction, and a finite endpoint changes which modes can occur.
- Q7. A certified algebraic-coefficient front end.** Implement exact factorization and algebraic modulus comparison for $P \in \mathbb{Q}[z]$, returning $\mathcal{R}(P)$ together with root-isolation certificates. Equal-modulus decisions, rather than approximate root clustering, are essential at the sharp boundary. The factor-level code shipped with Part II is a testable starting point, not that completed front end.
- Q8. Stability of the core under coefficient perturbation.** Determine the strata on which the degree and multiplicity pattern of $\mathcal{R}(P)$ are constant, and describe their jumps. Equal-modulus and positive-real-root collisions are natural boundary loci. A useful output would distinguish stable eliminations from eliminations destroyed by an arbitrarily small perturbation.
- Q9. Constructive positive envelopes and rational certificates.** The sufficiency proof uses an existentially large Λ and a small perturbation parameter. Produce explicit certified bounds in terms of root

separations, degrees, and coefficient heights. This would turn an existence construction into a predictable numerical and symbolic realization algorithm.

Q10. Lean formalization in separable stages. A practical sequence is: the annihilator ideal and difference-spectrum identity; bilateral recurrence decomposition; finite trigonometric returns and the positive-mean lemma; the necessary spectral test; the finite root-level core formula; then the analytic positive-envelope and conjugacy construction. The algebraic reduction and the finite classification can be audited independently before the analytic realization library is complete. (As of 29 September 2026 none of these stages has been started in ProveIt.)

These are proposed projects, not claims of unresolved status throughout all literature. The decreasing and regularity questions require a renewed bibliographic check before being presented as newly formulated problems. [Added 29 September 2026, batch 42: Part III's literature check for the decreasing question is recorded in Section 37; it too is not exhaustive.]

22 Conclusion of Part II

Removing one 1-factor converts the increasing iterative-equation problem into a problem about positive bilateral recurrences. At two different ends, different exponential envelopes can dominate. That permits even extreme multiplicities and arbitrary interior multiplicities, exactly where the circular theorem no longer supplies the answer.

The resulting classification is both necessary and sufficient, and its sufficiency is global: every permitted spectrum is realized by a bi-Lipschitz increasing homeomorphism. The universal core then identifies the exact amount of order reduction available from an arbitrary input polynomial, with one realization witnessing all retained multiplicities. The fixed-point theorems separate two dynamical regimes, and the analytic family shows that high nonreal multiplicity is compatible with uniformly controlled metric distortion.

The result closes the several-circle gap in Part I within the increasing whole-line setting. It does not assert a classification of decreasing maps, noninjective zero-constant equations, or every individual solution. Its mathematical claims are the stated and proved theorems; historical priority and independent formal verification remain separate questions. [Added 29 September 2026, batch 42: decreasing maps are classified in Part III.]

23 A finite Hankel certificate for minimality

Suppose a scalar sequence u_n is annihilated by a monic degree- d polynomial. If a polynomial of degree $e < d$ also annihilated it, then there would be a nontrivial linear relation among the first $e + 1$ columns of the Hankel matrix $(u_{i+j})_{0 \leq i, j \leq d-1}$: evaluate that recurrence at $i = 0, \dots, d - 1$. Hence a nonzero determinant proves minimal order d . This does not require numerical conditioning estimates. (Part I uses the same argument for (11) and (22).)

For the analytic example, the exact matrix is

$$\begin{pmatrix} 0 & 8/5 & 15/4 & 311/40 \\ 8/5 & 15/4 & 311/40 & 255/16 \\ 15/4 & 311/40 & 255/16 & 5131/160 \\ 311/40 & 255/16 & 5131/160 & 4095/64 \end{pmatrix},$$

with determinant $225/256$. Its recurrence polynomial is (98). The stored JSON contains the analogous exact matrix for the even-multiplicity example.

24 Source and novelty audit of Part II

Repository source. The inspected source is Part I, at `SetTheory/Cardinals/docs/reports/log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/article.tex` in `ProveIt` (<https://github.com/VladimirReshetnikov/ProveIt>), inspected on 28 September 2026. The GitHub content API returned blob SHA

ae6ea0aa0566316ba9ed06b1750c6e73275059da.

The retrieved conclusion explicitly leaves several-circle spectra unclassified. The manuscript does not reuse the circular counterexamples as its new result, and it does not claim that the repository’s unrefereed report is a formal certification.

External sources checked. The primary preprints [1, 2] were read for the formulation and scope of the order-reduction problem. Page 13 of [1], including Problem 6.1, was also visually inspected. Petrov’s original question and its displayed comments [3], including I. Bogdanov’s, were checked directly. (The manuscript cites `arXiv:1503.00570` without a version and `arXiv:1604.01287v2`; Part I’s bibliography entries, used here, give the same works.) Searches for broader iterative-spectrum classifications returned a mixture of relevant and irrelevant results; no exhaustive citation-database review was completed. Thus Part II establishes a concrete extension of Part I, not a certified first-in-literature claim.

Trust boundary. All universal statements in Part II rest on the written proofs. The rational scripts check explicit examples and finite consistency of the root-level rule. No specialized theorem from the repository or cited iterative-equation papers is used as an unproved premise; the argument uses standard real analysis and finite-dimensional linear algebra. No independent referee or proof assistant has checked the complete development. The manuscript’s author read the repository only; no files were committed or modified there. The manuscript’s own status note and source record are shipped verbatim as `02-spectral-core-STATUS.md` and `02-spectral-core-source_audit.md`.

Part III. Orientation-reversing spectra: the decreasing spectral core

Part III was added on 29 September 2026 in batch 42 of ProveIt’s incoming-report intake, from one manuscript. It answers the boundary that Parts I and II both leave open, “We have not classified decreasing homeomorphisms, for which consecutive iterates reverse order” (Section 11), and Part II’s first research question (Section 21, Q1), for continuous maps of $\mathbb{R}$ and equations with nonzero constant coefficient: Theorem 25.1 classifies the minimal iterative polynomials of decreasing homeomorphisms, Theorem 30.1 gives the exact universal reduction $C_{\downarrow}(P)$, and Corollary 30.3 shows that it needs one or two witnesses, exactly as Q1 anticipated. It adds a finite-smoothness rigidity theorem (Theorem 31.3) with no counterpart in Parts I–II. Zero constant coefficient and proper interval domains remain open. Parts I and II are unchanged except for the title page and dated pointers to this addition. Section 25 records the provenance and fixes the notation of the addition.

25 Provenance, conventions and the main results

25.1 One manuscript, one addition

Part III is batch-42 manuscript 02, an AI-assisted research draft prepared for Vladimir Reshetnikov, dated 29 September 2026 and titled *The Decreasing Spectral Core: Complete minimal spectra, finite-smoothness rigidity, and global polynomial conjugacy* (27 US Letter pages as delivered). It arrived as the archive `ProveIt_Orientation_Reversing_Spectra.zip` (inner directory `ProveIt_Orientation_Reversing_Spectra/`) in ProveIt commit 8315d24e3 and was placed in 3609d0473. It is pinned to ProveIt commit afb2d1227 (full hash in Section 37). At that commit this report contained Part I only, as the Git blob ae6ea0aa of `article.tex` (the blob that pins Part II; Section 12.1), so the manuscript’s statements about “the ProveIt report” refer to the text of Part I printed above. It read Part II not in the repository but as the batch-41 manuscript itself, a copy in Vladimir’s file library (`article(20260929-042518).tex`), and does not assert that this copy belongs to the pinned commit; Part II prints that manuscript. Its verifier, Makefile, recorded output and status note are shipped beside Parts I–II with the prefix `03-decreasing-core-`; its manuscript, PDF, delivery README and checksum ledger (`CHECKSUMS.sha256`, 8/8 verified) are not shipped and survive in the arrival commit.

The manuscript’s abstract. For a real polynomial $P(z) = \sum a_j z^j$, write $P[f] = \sum a_j f^{\circ j}$, with composition in place of pointwise powers. We classify all minimal iterative polynomials of decreasing homeomorphisms of $\mathbb{R}$. Apart from involutions, the nonunit roots must all lie strictly inside or all strictly outside the unit circle. On each extreme modulus circle, its negative real root must have maximal multiplicity; on a single circle that multiplicity must also be odd. These conditions, with optional simple factors at 1 and -1 , are sufficient, with globally bi-Lipschitz realizations analytic away from at most two points.

The classification gives an explicit root-by-root formula for the smallest polynomial annihilating every decreasing solution of $P[f] = 0$. Two solutions always suffice to witness its sharpness, and two are necessary precisely when both contracting and expanding spectral branches survive.

A second theorem is substantially more rigid. Every non-involutive C^∞ decreasing solution is globally of the form $f(c + H(t)) = c + H(at)$, where $a < 0$, $a \neq -1$, and H is a polynomial with $H(0) = 0$, $H'(0) = 1$, and $H' > 0$ on $\mathbb{R}$. In fact, an explicit finite C^N hypothesis suffices. Consequently these

solutions are automatically real analytic and algebraic, and their minimal roots are distinct real integer powers of one negative multiplier. We derive the exact smooth spectral core, exhibit unbounded smooth witness complexity, supply self-contained proofs and exact finite checks, and identify further regularity and formalization problems.

The manuscript contributes, in the order printed here: its statement of the target and the two main theorems (this section); a regularity complement to Part I’s homeomorphism lemma and a two-periodic limit lemma (Section 26); the two-cycle interval and the global separation of contracting and expanding spectra (Section 27); its use of Part II’s positive-recurrence theorem (Section 28); the classification of decreasing minimal spectra (Section 29); the universal decreasing core, its two sharp branches, the witness count and an affine rigidity test (Section 30); C^1 hyperbolicity, an explicit smoothness threshold and the polynomial-coordinate theorem (Section 31); smooth spectra, the smooth core and unbounded smooth witness counts (Section 32); examples (Section 33); its finite checks and a formalization route (Section 34); ten research questions (Section 35); its conclusion (Section 36); and its source, dependency and claim audit (Section 37). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- (a) The manuscript reproves, so as not to assume Parts I–II, the homeomorphism lemma, the annihilator ideal and cyclic composition space, the two-sided recurrence decomposition, orbitwise least common multiples, the simultaneous returns and positive means of finite trigonometric sequences, and, from Part II, positive admissibility with its necessity proposition and least-common-multiple lemma, the two-ended positive envelope and the positive analytic interpolation theorem. Its proofs are the same arguments as those printed in Parts I and II. They are printed once, there, and credited where Part III uses them (Sections 26.1 and 28). What the manuscript adds to them is printed: the regularity of the inverse (Lemma 26.1), the trigonometric limit statement (Lemma 26.2(iii)) and the two-periodic limit lemma (Lemma 26.3), which extends Part II’s Lemma 18.1.
- (b) Remarks 30.2 and 30.5 were added in the merge. The first expresses the decreasing core through Part II’s positive-recurrence core $\mathcal{K}$ and records a finite check of that identity made in the write phase; the second combines Part II’s and Part III’s rigidity theorems into an answer to Petrov’s linearity question for equations with nonzero constant coefficient.
- (c) Symbols were renamed to avoid collisions with Parts I–II and within the manuscript (Section 25.2); no normalization changed. The manuscript’s $C_\downarrow$ and C_∞ are kept. Its references to “the ProveIt report” and to “the increasing companion” now read “Part I” and “Part II”, and its file names are replaced by the shipped names.
- (d) The manuscript’s appendix (its source audit) is printed as the last section of Part III, so that Part I’s appendices keep their letters. Its unnumbered verification table stays unnumbered.

Status (manuscript 02 of batch 42). This is an AI-assisted, unrefereed research draft, not a Lean- or Rocq-verified development, and no declaration in ProveIt formalizes any statement of Parts I–III. The main classifications and the finite-smoothness theorem are proved below, rather than assumed from the repository; no theorem of an unverified repository manuscript is used as an unproved premise. Their extension beyond the inspected repository reports is explicit; a complete historical-priority claim is not made. Classical recurrence methods and the previously known two-branch phenomenon (Draga’s Theorem 3 and Remark 4 [2]) are

credited. The domain is the whole real line and $P(0) \neq 0$: Part III does not classify maps of arbitrary intervals, equations with zero constant coefficient, nonhomogeneous equations, higher-dimensional systems, or every individual continuous solution (it does classify every individual non-involutive solution in the stated sufficiently smooth class). The finite-smoothness threshold is sufficient; its optimality is not claimed. The finite computations check the algebraic rules on finite atlases; they are not proofs of the all-real theorems.

25.2 Notation across the three parts

Part I's and Part II's symbols keep their meanings in their parts. Table 4 fixes the meanings used in Part III; letters not listed (P , Q , μ_f , E , Ann , Fix , W , G , Λ as in Part II's envelope lemma, and $\mathcal{K}$) agree with Parts I–II. For a monic polynomial A and a root λ , $m_A(\lambda)$ is the multiplicity of λ in A (Part II's notation; the manuscript writes $\text{mult}_A(\lambda)$), and $\text{Roots}(A)$ is the set of distinct complex roots.

25.3 The research target and the contribution

Part I studies the obstruction to eliminating nonreal characteristic roots and classifies increasing minimal spectra supported on a single modulus circle. Its scope does not include a complete decreasing classification. Part II (the manuscript's "increasing companion") treats arbitrary-radius increasing spectra and explicitly excludes decreasing solutions. Part III addresses that remaining orientation, then goes beyond spectral occurrence to a global classification under smoothness assumptions.

The distinction is not a change of a sign in the increasing theorem. A decreasing homeomorphism has a unique fixed point, but its square may have an interval of fixed points. Every exterior orbit approaches a finite two-cycle in one time direction. This forces a global separation between contraction and expansion and accounts for two different universal reductions. After sufficient differentiability is imposed, the two-cycle interval collapses and all non-polynomial spectral behavior disappears.

The external background is the characteristic-root and order-reduction program, including Yang–Zhang [6], Draga–Morawiec [1], and Draga [2]. The use of bilateral orbit recurrences is classical. In particular, Draga's Theorem 3 eliminates one of two negative extreme roots straddling -1 , and his Remark 4 already explains why the result can be a disjunction of two lower-order equations. We do not claim that phenomenon or its quadratic example as new. Our target is the exact arbitrary-multiplicity classification, its sharp reduction formula, and the separate finite-smoothness rigidity theorem.

25.4 Conventions and main statements

Throughout Part III, $P \in \mathbb{R}[z]$ is nonzero and $P(0) \neq 0$. Constant nonzero polynomials are permitted as input equations, although they have no solutions. Iterates are denoted by $f^{\circ j}$, with $f^{\circ 0} = \text{id}$; negative iterates use the inverse. A map is *decreasing* here if it is strictly decreasing. Its *minimal iterative polynomial* μ_f is the monic generator of

$$\text{Ann}(f) = \{Q \in \mathbb{R}[z] : Q[f] = 0\}.$$

The generator exists whenever f satisfies a nonzero polynomial iterative equation. All multiplicities are complex-root multiplicities, with conjugate roots counted separately in polynomial degrees.

For a polynomial R with no root on the unit circle, say that R is *negative-admissible* when, on its smallest and largest modulus circles, the negative real root has multiplicity at least as large as that of every other root on the same circle. If there is just one modulus circle, require this negative-root multiplicity to be odd. There is no restriction on multiplicities on strictly intermediate circles.

Table 4: Symbols of Part III, with the tempting false reading beside the true one.

Symbol	Meaning in Part III	Not to be confused with
$C_{\downarrow}(P)$	the decreasing spectral core: the least universal annihilator of all continuous decreasing solutions (Theorem 30.1)	Part II's <i>increasing</i> core $\mathcal{R}(P)$ (87)
$C_{\infty}(P)$ negative-admissible	the smooth decreasing core (Theorem 32.2) Definition in Section 25.4	$C_{\downarrow}(P)$, which it divides Part II's positive admissibility: R is negative-admissible iff $R^{\#}$ is positive-admissible
$A^{\#}$	the sign reflection $(-1)^{\deg A} A(-z)$ (110)	—
$\varepsilon_+, \varepsilon_-$	the exponents of $z - 1$ and $z + 1$; in (119), $\mathbf{1}_{P(1)=0}$ and $\mathbf{1}_{P(-1)=0}$ (<i>renamed</i> from the manuscript's ε, δ)	Part II's $\delta(A)$, which is ε_+ (not ε_-); Part I's perturbation coefficients ε_j
U	the unit core $(z - 1)^{\varepsilon_+} (z + 1)^{\varepsilon_-}$	—
$R; R_S, R_<, R_>$	the nonunit part of μ (Theorem 25.1); the side cores of P (Section 30); R_m is the example (143)	Part II's $R = \mathcal{R}(P)$ in its rigidity proof; Part II's factors $\mu_<, \mu_>$ of μ (here $R_< = \mathcal{K}(P_<)^{\#}$, Remark 30.2)
A_S r_-, r_+	the set of negative-anchor radii on the side S in Section 30: least and greatest negative anchors, which are the extreme root moduli of R_S	a polynomial A (Part II); Part I's function $A(t)$ $r_-(P), r_+(P)$ below
$r_-(P), r_+(P)$	least and greatest root moduli of P (<i>renamed</i> from $r(P), R(P)$)	Part II's core $\mathcal{R}(P)$
κ	the largest odd integer $\leq m_P(-r)$ (120) (<i>renamed</i> from h), Part II's κ after reflection	Part I's k , where $m_r = 2k + 1$
ω	a lower bound for root moduli (Lemma 26.3(iii)); $\min_{\lambda} \lambda ^{\sigma}$ in (128) (<i>renamed</i> from κ)	κ above
$\Theta_P(a), \Theta(P)$	the smoothness threshold ratios (126) (<i>renamed</i> from $D_P(a), D(P)$)	Part II's difference spectrum $\mathcal{DP}$; Part I's dilations D_2, D_8
N	a differentiability order with $N \geq 1, N > \Theta(P)$ (127)	Part I's return times N_k
E_g	right composition $v \mapsto v \circ g$ on V (<i>renamed</i> from T)	a trigonometric sequence $T(n)$; Part I's translation T
$c; \alpha \leq \beta$	the unique fixed point of f ; $\text{Fix}(f^{\circ 2}) = [\alpha, \beta]$	Part II's endpoint c and range $(\alpha, \beta) = h(\mathbb{R})$
g	$f^{\circ 2}$ in Section 27 and Proposition 31.1; the centred map $f(c + x) - c$ in Lemma 31.2 and Theorem 31.3	Part II's positive coordinate g , which is w here
w, h	positive analytic exponential polynomial (Part II's g) and its integral, a bijection $\mathbb{R} \rightarrow (0, \infty)$	Part II's h onto an interval
$a; \sigma$	the multiplier $f'(c) < 0$ (or a negative root); the contracting time direction	Part II's rate $a = \log r_-$; Part I's coefficients a_j
b	the half-width of the central interval $[-b, b]$; in Lemma 31.2, $ a ^{\sigma} < 1$	Part II's $b_{\pm}$
$H; K, h_k$	the normalized polynomial coordinate; its support and coefficients	Part I's conjugating map H (14); Part II's modes H_{λ}
$K_P(a), J_P(a),$ $M_P(a), B_a, B_{\text{inv}},$ $\mathcal{B}(P)$	the smooth branch data (134)–(136)	Part II's positive-recurrence core $\mathcal{K}(B)$
F	an odd decreasing realization $F(\pm h(t)) = \mp h(t + 1)$, and other explicit maps	Part I's cubic counterexample F (Theorem 2.1)
witness	a member of a witness family, whose minimal polynomials have the core as least common multiple	Part II's single witness: here one or two suffice continuously, and unboundedly many may be needed smoothly

Theorem 25.1 (Continuous spectral classification, preview). *A monic nonconstant real polynomial μ with $\mu(0) \neq 0$ is the minimal polynomial of a decreasing homeomorphism of $\mathbb{R}$ if and only if either*

$$\mu = z + 1 \quad \text{or} \quad \mu = (z - 1)(z + 1),$$

or

$$\mu = (z - 1)^{\varepsilon_+}(z + 1)^{\varepsilon_-}R, \quad \varepsilon_+, \varepsilon_- \in \{0, 1\}, \quad (104)$$

where R is nonconstant, negative-admissible, and all its roots have modulus less than 1, or all have modulus greater than 1. Every permitted μ has a globally bi-Lipschitz realization, real analytic outside at most two points.

Theorem 25.2 (Smooth rigidity, preview). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^∞ decreasing solution of $P[f] = 0$. Either $f^{\circ 2} = \text{id}$, or there are unique $c \in \mathbb{R}$, $a < 0$ with $a \neq -1$, and $H \in \mathbb{R}[t]$ such that*

$$H(0) = 0, \quad H'(0) = 1, \quad H'(t) > 0 \quad (t \in \mathbb{R}), \quad f(c + H(t)) = c + H(at). \quad (105)$$

Here c is the fixed point and $a = f'(c)$. If $H(t) = \sum_{k \in K} h_k t^k$ with every $h_k \neq 0$, then

$$\mu_f(z) = (z - 1)^{\mathbf{1}_{c \neq 0}} \prod_{k \in K} (z - a^k), \quad 1 \in K, \quad \max K \text{ odd}. \quad (106)$$

Moreover, a computable finite differentiability threshold depending only on the root moduli of P can replace C^∞ .

The full proofs appear in Sections 29 and 31. The threshold is stated explicitly in Theorem 31.3. Sections 30 and 32 turn these theorems into exact universal reductions.

25.5 What is inherited and what is developed here

The positive-recurrence criterion used in Section 28 is the mechanism of Part II; the manuscript reproves it because the decreasing construction needs its continuous positive interpolation, not merely a necessary condition. Neither its reuse nor the orientation-reversed cubic example is the principal claim. The substantive continuation is the complete decreasing classification, including the unit-root and two-cycle geometry, the maximal reduction branches with arbitrary multiplicities, and the global polynomial-coordinate theorem under an explicit finite regularity assumption. No theorem from an unverified repository manuscript is used as an unproved premise.

26 Linear algebra behind a nonlinear equation

26.1 What is taken from Parts I and II

The manuscript opens by reproving facts that Part I proves by the same arguments; they are used here in Part I's form.

- (1) Every continuous solution $f : \mathbb{R} \rightarrow \mathbb{R}$ of $P[f] = 0$ is a strictly monotone homeomorphism (Lemma 4.3).
- (2) $\text{Ann}(f)$ is an ideal, generated by μ_f , because $(zQ)[f] = Q[f] \circ f$ is right composition (Lemma 4.2, (25)). One must not replace it by left composition and distribute f over a sum; that invalid maneuver would destroy the linear-algebra argument.

- (3) If $A(0) \neq 0$, every complex bilateral sequence annihilated by $A(E)$, where $Eu(n) = u(n+1)$, has a unique expansion $u(n) = \sum_{\lambda \in \text{Roots}(A)} p_\lambda(n) \lambda^n$ with $\deg p_\lambda < m_A(\lambda)$, and its minimal recurrence polynomial is $\prod_{p_\lambda \neq 0} (z - \lambda)^{1+\deg p_\lambda}$; real sequences have conjugate coefficient polynomials at conjugate roots (Lemma 4.4, (27)).
- (4) The scalar orbit $u_x(n) = f^{\circ n}(x)$ is such a sequence. A polynomial annihilates all orbits if and only if it annihilates f , by evaluation at $n = 0$. Hence μ_f is the least common multiple of the orbit minimal polynomials, and only finitely many orbits are needed (Lemma 4.5).

As in Part II, let E act on real-valued functions by $Ev = v \circ f$. The vector space

$$V_f = \text{span}_{\mathbb{R}}\{\text{id}, f, f^{\circ 2}, \dots\}$$

is finite dimensional when $P[f] = 0$. Right composition is linear on this space, even though f need not be a linear map of $\mathbb{R}$. It is invertible by (107) below, and id is a cyclic vector. The minimal polynomial of $E|_{V_f}$ equals μ_f , and $\mu_f(0) \neq 0$.

The manuscript's version of the homeomorphism lemma adds regularity of the inverse.

Lemma 26.1. *Every continuous solution $f : \mathbb{R} \rightarrow \mathbb{R}$ of $P[f] = 0$ is a strictly monotone homeomorphism. If f is C^k , $1 \leq k \leq \infty$, then its inverse is C^k and $f'(x) \neq 0$ everywhere. The analogous assertion holds for real analytic f .*

Proof. The first sentence is Lemma 4.3; the manuscript gives the same proof. Write $P(z) = \sum_{j=0}^d a_j z^j$, $a_0 a_d \neq 0$. Substitution of $x = f^{-1}(y)$ gives

$$f^{-1}(y) = -\frac{1}{a_0} \sum_{j=1}^d a_j f^{\circ(j-1)}(y). \quad (107)$$

This proves the regularity assertion. Differentiating $f^{-1} \circ f = \text{id}$ proves that f' is nowhere zero. The constant-polynomial case has no solution and is vacuous. $\square$

26.2 Trigonometric returns and two-periodic limits

Lemma 26.2 (Elementary trigonometric facts). *Let $T(n) = \sum_{\zeta \in S} c_\zeta \zeta^n$, with distinct $|\zeta| = 1$. There are integers $N_j \rightarrow +\infty$ such that $\zeta^{N_j} \rightarrow 1$ for all $\zeta \in S$. Consequently:*

- (i) *if T is real and $\liminf_{n \rightarrow \infty} T(n) \geq 0$, then $T(n) \geq 0$ for every integer n ;*
- (ii) *a nonzero nonnegative real T has a strictly positive coefficient at frequency 1;*
- (iii) *if $T(n)$ has a limit as $n \rightarrow \infty$, then T is constant.*

Proof. The returns are Lemma 5.1, and (i) and (ii) are Lemma 14.1 (that is, Lemma 5.1 with the proof of Theorem 5.3, and Lemma 5.2). For (iii), fix ℓ ; then $T(N_j + \ell) \rightarrow T(\ell)$, and the left side tends to the limit, so $T(\ell)$ equals the limit for every ℓ . (Equivalently, T minus its limit is a unit-circle exponential polynomial tending to zero, hence zero by Corollary 5.4.) The manuscript proves all three by the same compactness and Cesàro-mean arguments; the same return argument works in negative time. $\square$

Lemma 26.3 (Boundedness, convergence, and spectral size). *The following hold for a nonzero sequence of the form (27).*

- (i) If it is bounded for all $n \in \mathbb{Z}$, all active roots are simple and have modulus 1.
- (ii) If it has finite limits along the even and odd subsequences as $n \rightarrow +\infty$, its active roots have modulus less than 1, except possibly simple roots 1 and -1 . Their contribution is exactly the limiting two-periodic sequence. For $n \rightarrow -\infty$, replace “less” by “greater”.
- (iii) If every active root has modulus at least $\omega > 0$, then $u(n) = O(\rho^n)$ with $0 < \rho < \omega$ is impossible as $n \rightarrow +\infty$.

Proof. Let r be the largest active modulus and d the largest polynomial degree on that circle. Then

$$u(n) = r^n n^d (T(n) + o(1)),$$

where T is a nonzero finite trigonometric sequence. Such a T is bounded away from zero along a subsequence: its positive mean square precludes convergence to zero. This rules out exponential growth and then polynomial growth when boundedness is required. Applying the same argument to $u(-n)$ proves (i), and proves the growth exclusions in (ii). The remaining unit-circle component has convergent even and odd subsequences. Group its frequencies after squaring; Lemma 26.2(iii) shows that only $\lambda^2 = 1$ can remain. Considering both parities rules out cancellation between λ and $-\lambda$ with $\lambda^2 \neq 1$. This proves (ii). The same leading-term argument proves (iii). $\square$

Part II’s Lemma 18.1 is the case of (ii) in which the whole sequence has one limit; then only a simple root 1 can remain on the unit circle. The two-periodic version is what decreasing maps need.

27 Two-cycles force a global spectral separation

Theorem 27.1 (The two-cycle interval). *Let f be a decreasing solution of $P[f] = 0$, and put $g = f^{\circ 2}$. The map f has a unique fixed point c . Either $g = \text{id}$, or*

$$\text{Fix}(g) = [\alpha, \beta], \quad -\infty < \alpha \leq c \leq \beta < \infty, \quad f(\alpha) = \beta, \quad (108)$$

with $f(\beta) = \alpha$. On (β, ∞) , either $g(x) > x$ for every x , or $g(x) < x$ for every x . In exactly one common time direction, every exterior orbit tends to the endpoint two-cycle.

Proof. The continuous function $f(x) - x$ is strictly decreasing from $+\infty$ to $-\infty$, so it has exactly one zero c . Suppose $p < q$ are fixed by g and $p < x < q$. All even iterates of x remain in $[p, q]$, in both time directions, and all odd iterates remain in $[f(q), f(p)]$. Thus the whole bilateral orbit is bounded. By Lemma 26.3(i), it is a finite trigonometric sequence. The even subsequence $g^{\circ n}(x)$ is monotone in n , because g is increasing, and is bounded. It therefore converges, and Lemma 26.2(iii), applied to the squared frequencies, shows that it is constant. Hence $g(x) = x$.

This proves that $\text{Fix}(g)$ is a nonempty closed interval. It is invariant under f . If it were unbounded above, its image under the decreasing homeomorphism f would be unbounded below, and conversely. Thus a proper such interval is bounded on both sides, proving (108). On its right complement the continuous function $g(x) - x$ has no zero, hence a fixed strict sign. If that sign is positive, $g^{\circ n}(x) \rightarrow \beta$ as $n \rightarrow -\infty$, and tends to $+\infty$ as $n \rightarrow +\infty$; any other finite limit would be a further fixed point. For the negative sign, reverse time. Odd iterates tend to α at the finite end. The left ray is the image of the right ray under f , so its time direction is the same. $\square$

Corollary 27.2 (Unit roots and one-sided nonunit spectrum). *For a non-involutive decreasing solution, all nonunit roots of μ_f lie on one side of the unit circle: all have modulus less than 1, or all have modulus greater than 1. The only possible unit roots are simple 1 and simple -1 . Moreover,*

$$-1 \in \text{Roots}(\mu_f) \iff \alpha < \beta. \quad (109)$$

If $\alpha = \beta = c$, the root 1 occurs if and only if $c \neq 0$.

Proof. Every exterior orbit has a finite two-periodic limit in the common time direction from Theorem 27.1. Apply Lemma 26.3(ii). Every interior orbit is exactly two-periodic. Taking the finite least common multiple of their minimal polynomials preserves these restrictions and proves the global, rather than merely orbitwise, separation.

When $\alpha < \beta$, the endpoint orbit itself has a nonzero $(-1)^n$ component. When the interval is a singleton, every orbit has constant limit c and no $(-1)^n$ component. In that case the constant component is exactly c , proving the last assertion. $\square$

Remark 27.3. For a genuine two-cycle interval, the presence of the root 1 is not determined just by the fixed point. A nonlinear involution on the interval may have nonconstant pair averages even when $c = 0$. The equivalence between a simple root 1 and a nonzero fixed point is asserted only when the interval is a singleton.

The periodic cases are now completely separated. If $f^{\circ 2} = \text{id}$, then μ_f divides $z^2 - 1$. It cannot be $z - 1$, since f is decreasing. Thus it is either $z + 1$, which forces $f = -\text{id}$, or $z^2 - 1$. Conversely, reflections $x \mapsto -x$ and $x \mapsto 2c - x$, $c \neq 0$, realize these two polynomials.

Corollary 27.4 (Bounded-interval version). *A decreasing homeomorphism of a bounded interval that satisfies a nonzero polynomial iterative equation is an involution.*

Proof. Every bilateral orbit is bounded. The bounded-orbit argument in the proof of Theorem 27.1 makes each even subsequence constant. This does not require the interval to be closed. $\square$

28 Positive bilateral recurrences and their realization

The manuscript's Section 4 reproves Part II's positive-recurrence theory with Part II's statements and arguments; it is used here in Part II's form, with one change of letter: the positive exponential polynomial that Part II calls g is called w in Part III, because g denotes $f^{\circ 2}$ and a centred map here.

- (1) *Positive admissibility* is Part II's Definition 12.1: the positive real root on each extreme modulus circle has maximal multiplicity on that circle, and it is odd when there is only one circle.
- (2) The minimal polynomial of a nonzero nonnegative bilateral recurrence sequence is positive-admissible, and a least common multiple of finitely many positive-admissible polynomials is positive-admissible (Proposition 14.2 and Lemma 15.1; the manuscript states them as one proposition, with the same proofs).
- (3) The two-ended positive envelope G and weight W of Lemma 14.3 (the manuscript's large constant T is Part II's Λ).
- (4) *Exact positive interpolation* (Theorem 14.4): a monic nonconstant real B with $B(0) \neq 0$ is positive-admissible if and only if it is the minimal polynomial of a strictly positive bilateral recurrence sequence,

and every such B is the exact unit-shift minimal polynomial of a positive real-analytic exponential polynomial $w : \mathbb{R} \rightarrow (0, \infty)$ with $0 < m \leq w(t+1)/w(t) \leq M < \infty$ (Part II's (78)), comparable above and below to the weight W of (77), and built from the modes (79).

Remark 28.1 (A sampling pitfall). Exact minimality is a statement about the continuous coordinate, or about a nonsingular sampling phase. A sine term created by integration can vanish at every integer without being the zero function. Neither integer samples alone nor positive numerical samples establish the continuous minimality or positivity required here. (This is the manuscript's restatement of the caution printed after Theorem 14.4.)

29 Complete classification of decreasing minimal spectra

For a monic polynomial A , define its sign reflection by

$$A^\#(z) = (-1)^{\deg A} A(-z). \quad (110)$$

Its roots are the negatives of the roots of A , with unchanged multiplicities. Thus negative-admissibility of R is precisely positive-admissibility of $R^\#$.

29.1 Necessity: positive distances from the limiting two-cycle

Necessity in Theorem 25.1. The involution case was settled after Corollary 27.2. Suppose f is not an involution. Write the unit part of its minimal polynomial as $(z-1)^{\varepsilon_+}(z+1)^{\varepsilon_-}$; the exponents are at most one by that corollary. Its nonunit part R is nonconstant. Indeed, if there were no nonunit roots, μ_f would divide $z^2 - 1$ and f would be an involution. All roots of R lie on the same side of the unit circle.

Let $x > \beta$, where $[\alpha, \beta] = \text{Fix}(f^{\circ 2})$, and set $b_n = \beta$ for even n , $b_n = \alpha$ for odd n . Then

$$v_x(n) = (-1)^n (f^{\circ n}(x) - b_n) > 0 \quad (n \in \mathbb{Z}). \quad (111)$$

The limiting unit component of the orbit is exactly b_n , by Lemma 26.3. Hence subtraction removes all its unit roots and no nonunit root. Multiplication by $(-1)^n$ negates the remaining roots. Every left exterior orbit is a shift of a right exterior orbit, and every interior orbit is two-periodic. Consequently the least common multiple of the minimal polynomials of the v_x is $R^\#$, and finitely many x suffice to attain it.

Part II's Proposition 14.2 and Lemma 15.1 imply that $R^\#$ is positive-admissible. Therefore R is negative-admissible. This proves all necessary conditions. $\square$

29.2 Sufficiency: an odd analytic construction on two rays

Lemma 29.1 (Integrating a one-sided spectrum). *Let B be positive-admissible, with all roots of modulus greater than 1, or all of modulus less than 1. There is a positive real-analytic bijection $h : \mathbb{R} \rightarrow (0, \infty)$, increasing in the former case and decreasing in the latter, whose exact unit-shift polynomial is B . It satisfies $|h'(t)| = w(t)$ for a positive w with bounded ratios $w(t+1)/w(t)$ as in (78).*

Proof. Use w from Theorem 14.4 (Part II's g), and define

$$h(t) = \begin{cases} \int_t^t w(s) ds, & \text{all root moduli exceed 1,} \\ \int_t^{-\infty} w(s) ds, & \text{all root moduli are below 1.} \end{cases} \quad (112)$$

The envelopes show integrability at the displayed finite end and nonintegrability at the other end. Thus h is a bijection as stated, with derivative nowhere zero.

Every continuous exponent s used in constructing w has nonzero real part. For a polynomial p , a primitive of $e^{st} p(t)$ is Part II's (80). This preserves its nonzero leading degree. The primitive vanishes at the convergent end in (112), so no constant mode is added. The exact minimal polynomial is still B . $\square$

Sufficiency and regularity in Theorem 25.1. Let R satisfy the nonunit conditions. Apply Lemma 29.1 to $B = R^\#$, and define $F(0) = 0$ and

$$F(h(t)) = -h(t+1), \quad F(-h(t)) = h(t+1). \quad (113)$$

The same-direction change of $h(t)$ and $h(t+1)$ makes F strictly decreasing on each ray, and the signs make it strictly decreasing across zero. The finite-end limit makes it continuous there, and the infinite-end limit makes it onto. Its iterates satisfy

$$F^{\circ n}(\pm h(t)) = \pm(-1)^n h(t+n) \quad (n \in \mathbb{Z}). \quad (114)$$

For every polynomial Q , the equation $Q[F] = 0$ on the positive ray is therefore equivalent to $Q(-E)h = 0$. Since the exact shift polynomial of h is $R^\#$, the exact iterative polynomial of F is R .

The inverse function theorem shows analyticity on both open rays, and

$$|F'(\pm h(t))| = \frac{w(t+1)}{w(t)}. \quad (115)$$

This is bounded above and away from zero. The mean value theorem on each ray and continuity at zero extend the same Lipschitz bounds across zero. Thus F is globally bi-Lipschitz.

To add one factor $z+1$, choose $b > 0$, use $F(x) = -x$ on $[-b, b]$, and replace (113) by

$$F(b+h(t)) = -b-h(t+1), \quad F(-b-h(t)) = b+h(t+1). \quad (116)$$

The exterior orbit becomes $\pm(-1)^n(b+h(t+n))$. Since R has no unit root, its new $(-1)^n b$ component adds exactly the simple factor $z+1$. The map is still globally bi-Lipschitz, with bounds incorporating the slope 1 on the central interval, and it is analytic away from the two endpoints.

Finally, to add one factor $z-1$, translate any of these odd maps:

$$F_c(x) = c + F(x-c), \quad c \neq 0. \quad (117)$$

For a polynomial Q , $Q[F_c](x) = cQ(1) + Q[F](x-c)$. The second summand is odd as a function of $x-c$. It can cancel the first everywhere only when both are zero. Thus $\mu_{F_c} = \text{lcm}(\mu_F, z-1)$, which adds exactly one copy of $z-1$. The pure involution cases are realized by reflections. All cases of (104) are now constructed. $\square$

Corollary 29.2 (Existence criterion). *There is a decreasing continuous solution of $P[f] = 0$ if and only if P has a negative real root.*

Proof. Every permitted minimal polynomial has a negative real root. Conversely, if $a < 0$ and $P(a) = 0$, the linear map $f(x) = ax$ is a solution. $\square$

(Added in the merge.) Together with Part II's Theorem 16.3(ii), which gives increasing solutions exactly when P has a positive real root, this says that $P[f] = 0$ has a continuous solution on $\mathbb{R}$ if and only if P has a real root, every continuous solution being monotone by Lemma 4.3.

Remark 29.3 (Not every formal least common multiple is realizable). The least common multiple of two decreasing minimal spectra need not itself be a decreasing minimal spectrum. A contracting and an expanding nonunit spectrum cannot coexist in one map. This is the precise point at which the increasing and decreasing universal reductions differ.

30 The exact universal decreasing core

The objective is a monic polynomial $C_{\downarrow}(P)$ such that, for every $Q \in \mathbb{R}[z]$,

$$\left[\begin{array}{l} P[f] = 0 \implies Q[f] = 0 \text{ for every} \\ \text{continuous decreasing } f : \mathbb{R} \rightarrow \mathbb{R} \end{array} \right] \iff C_{\downarrow}(P) \mid Q. \quad (118)$$

When there are no decreasing solutions, this is the vacuous property and $C_{\downarrow}(P) = 1$. By Corollary 29.2, this happens exactly when P has no negative real root.

30.1 A closed root-by-root construction

Assume henceforth that at least one negative root exists. Set

$$\varepsilon_+ = \mathbf{1}_{P(1)=0}, \quad \varepsilon_- = \mathbf{1}_{P(-1)=0}, \quad U(z) = (z-1)^{\varepsilon_+}(z+1)^{\varepsilon_-}. \quad (119)$$

Construct two polynomials $R_<, R_>$, independently, as follows. For the selected side $S = (0, 1)$ or $S = (1, \infty)$, let

$$A_S = \{r \in S : P(-r) = 0\}$$

be its set of *negative-anchor radii*.

No negative anchor. If A_S is empty, set $R_S = 1$.

One negative anchor. If $A_S = \{r\}$, let

$$\kappa = 2 \left\lfloor \frac{m_P(-r) - 1}{2} \right\rfloor + 1, \quad (120)$$

the largest odd integer not exceeding $m_P(-r)$. Keep exactly the roots on $|z| = r$, with multiplicities

$$m_{R_S}(\lambda) = \min\{m_P(\lambda), \kappa\}. \quad (121)$$

Discard all other roots on this side.

Two or more negative anchors. Let $r_- = \min A_S$ and $r_+ = \max A_S$. Discard roots outside the closed annulus $r_- \leq |z| \leq r_+$. Inside its open part, keep every multiplicity unchanged. On its boundary circles set

$$m_{R_S}(\lambda) = \begin{cases} \min\{m_P(\lambda), m_P(-r_-)\}, & |\lambda| = r_-, \\ \min\{m_P(\lambda), m_P(-r_+)\}, & |\lambda| = r_+. \end{cases} \quad (122)$$

These instructions define $R_{<} := R_{(0,1)}$ and $R_{>} := R_{(1,\infty)}$ completely. In particular, nonreal unit roots are always discarded, and the multiplicities at ± 1 are clipped to one by (119).

Theorem 30.1 (Closed universal reduction). *With the conventions above,*

$$\boxed{C_{\downarrow}(P) = UR_{<}R_{>}} \quad (123)$$

when decreasing solutions exist, and $C_{\downarrow}(P) = 1$ otherwise. This polynomial has property (118) and divides P up to its nonzero leading scalar.

Proof. Fix one side of the unit circle. Every nonconstant negative-admissible divisor of P on that side must have negative roots on both of its extreme circles. If there is one available negative anchor, the divisor must have just that circle; its negative-root multiplicity is odd and at most κ , so (121) is an upper bound for all its multiplicities. The polynomial defined there itself satisfies the admissibility conditions.

With at least two available anchors, no root outside $[r_-, r_+]$ can occur in such a divisor. A root on one of these outer boundary circles, if used, must be bounded in multiplicity by the negative root of that circle. Thus (122) bounds all permitted divisors. In the open annulus there is no universal multiplicity loss: the proposed R_S includes all roots there and uses the two outer negative anchors to satisfy its only extreme-circle conditions. It is itself negative-admissible. Therefore R_S is both an upper bound for, and a member of, the set of possible nonunit minimal spectra on the selected side.

Theorem 25.1 now implies that every decreasing minimal polynomial dividing P divides $UR_{<}R_{>}$. Conversely, whenever $R_S \neq 1$, that theorem realizes UR_S as an exact minimal polynomial. If both side polynomials are 1, existence forces $\varepsilon_- = 1$, and U is realized by an involution. Thus the least common multiple of attainable minimal polynomials is exactly (123). A polynomial Q annihilates every solution exactly when it is a multiple of each attainable minimal polynomial. This is (118). $\square$

Remark 30.2 (Added in the merge: the decreasing core through Part II's positive core). Let $P_{<}$ and $P_{>}$ be the monic factors of P formed from its roots of modulus less than and greater than 1 (Part II's $\mu_{<}, \mu_{>}$ applied to P), and let $\mathcal{K}$ be Part II's positive-recurrence core (Section 16), with $\mathcal{K}(1) = 1$ and $1^\# = 1$. If P has a negative real root, then

$$R_{<} = \mathcal{K}(P_{<}^\#)^\#, \quad R_{>} = \mathcal{K}(P_{>}^\#)^\#, \quad C_{\downarrow}(P) = (z-1)^{\varepsilon_+}(z+1)^{\varepsilon_-} \mathcal{K}(P_{<}^\#)^\# \mathcal{K}(P_{>}^\#)^\#.$$

Indeed the negative anchors on the side S are exactly the positive roots of $P_S^\#$, with $m_{P_S^\#}(r) = m_P(-r)$, and the three rules above are (85) (with the same κ) and (86) for $B = P_S^\#$, read back through the reflection. Compare Part II's $\mathcal{R}(P) = (z-1)^{\delta(P)} \mathcal{K}(\mathcal{D}P)$: the increasing core applies $\mathcal{K}$ once, to all of $\mathcal{D}P$ across the unit circle, whereas the decreasing core applies it to the two sides separately, keeps at most simple roots ± 1 and discards the other unit roots. The two separate applications are why two witnesses may be needed (Corollary 30.3). As a finite check made in the write phase, the closed core of the shipped verifier (code/03-decreasing-core-verify.py) was compared with the displayed formula evaluated by Part II's positive_core (code/02-spectral-core-spectral_core.py) on every profile of the four continuous atlases of Section 34 that has a negative real root: 2,552 of the 2,693 profiles, with no disagreement. The comparison script is not shipped; this description specifies it.

30.2 Two sharp branches, and the minimum witness count

A *witness family* for a universal core consists of solutions of the original equation whose minimal polynomials have that core as their least common multiple. Such a family certifies that no further factor or multiplicity can be deleted universally.

Corollary 30.3 (The sharp disjunction). *If $R_< \neq 1$ and $R_> \neq 1$, every continuous decreasing solution of $P[f] = 0$ satisfies at least one of*

$$(UR_<)[f] = 0 \quad \text{or} \quad (UR_>)[f] = 0. \quad (124)$$

Both maximal branch polynomials are attained by globally bi-Lipschitz solutions. Exactly two witnesses are required for $C_\downarrow(P)$. If solutions exist and at most one side survives, exactly one witness suffices. If there are no solutions, the empty family witnesses the constant core 1.

Proof. The nonunit roots of any one solution lie on a single side of the unit circle. Its unit roots divide U , and its nonunit part divides the corresponding side core. This proves the disjunction, including involutions, which satisfy both branch equations when both are present. The realization theorem supplies both maximal branches. Their least common multiple is the core. A single witness in the two-sided case would have a minimal polynomial containing nonunit roots on both sides, contradicting Corollary 27.2. The remaining cases follow from the same realization construction. $\square$

The two-branch phenomenon in this corollary includes the classical example discussed by Draga [2, Remark 4]. Its role here is a benchmark for a fully explicit maximum, not a new discovery of the existence of alternative reductions. The additional content is that all roots, all multiplicities, both maximal branches, and the exact minimum witness count are determined simultaneously.

30.3 An exact affine-rigidity test

Theorem 30.4 (When every decreasing solution is affine). *Assume P has a negative real root, and use (119). Every continuous decreasing solution of $P[f] = 0$ is affine if and only if one of the following mutually exclusive conditions holds:*

- (i) $\varepsilon_- = 1$, $\varepsilon_+ = 0$, and $R_< = R_> = 1$;
- (ii) $\varepsilon_- = 0$, and each nonconstant one of $R_<, R_>$ is linear.

Under these conditions, every solution is homogeneous linear if and only if $\varepsilon_+ = 0$.

Proof. In (i), the universal core is $z + 1$, so $f = -\text{id}$. In (ii), any nonconstant branch has the form $(z - 1)^{\varepsilon_+}(z - a)$, $a < 0$, $a \neq -1$. There is no two-cycle interval, since -1 is absent. If the fixed point is c , every orbit's unit component is the constant c . Centering the map removes it, leaving only the root a with multiplicity one. Thus $f(x) = c + a(x - c)$. If $\varepsilon_+ = 0$, the fixed-point equation forces $c = 0$.

Conversely, if $\varepsilon_- = 0$ and a side core has degree at least two, its exact realization cannot be affine: an affine decreasing map with slope $a \neq -1$ has minimal polynomial $z - a$ or $(z - 1)(z - a)$. If $\varepsilon_- = 1$ and a nonunit side survives, its realization with a nondegenerate two-cycle interval is non-affine. Finally, when $\varepsilon_- = \varepsilon_+ = 1$, the polynomial $z^2 - 1$ divides P and admits nonlinear decreasing involutions. For instance,

conjugating reflection by the nonodd increasing polynomial $H(t) = t + t^2/4 + t^3$ gives one; its derivative is positive, as checked in Section 33.

If $\varepsilon_+ = 1$, any negative root a gives nonzero translations of $x \mapsto ax$ with minimal polynomial dividing $(z - 1)(z - a)$, or translated reflections when $a = -1$. Hence not all solutions are homogeneous linear. This proves the final assertion. $\square$

Remark 30.5 (Added in the merge: Petrov's question with nonzero constant coefficient). Let $P(0) \neq 0$. Every continuous solution of $P[f] = 0$ on $\mathbb{R}$ is increasing or decreasing (Lemma 4.3), so Part II's Theorem 17.1 and Theorem 30.4 together decide Petrov's linearity question [3] for such P . Every continuous solution is affine if and only if (a) P has no positive root, or $\mathcal{R}(P) = z - r$ with $r > 0$, or $\mathcal{R}(P) = (z - 1)^2$; and (b) P has no negative root, or condition (i) or (ii) of Theorem 30.4 holds. Every continuous solution is homogeneous linear if and only if (a') P has no positive root or $\mathcal{R}(P) = z - r$ with $r > 0$; and (b') P has no negative root, or (i) or (ii) holds with $\varepsilon_+ = 0$. This combination is immediate and is not claimed as a separate result of either manuscript. Equations with $P(0) = 0$, whose solutions need not be injective, remain open.

31 Finite-smoothness rigidity and global polynomial conjugacy

The bi-Lipschitz realization theorem permits nonreal roots and repeated roots of arbitrarily large multiplicity. Sufficient differentiability at the finite end of the orbits eliminates both phenomena. The crucial point is that this is a global theorem, not just a local linearization. We prove it directly from the finite-dimensional composition space.

31.1 One derivative already rules out a two-cycle interval

Proposition 31.1 (C^1 hyperbolicity). *Let f be a C^1 decreasing solution of $P[f] = 0$ that is not an involution. Then*

$$\text{Fix}(f^{\circ 2}) = \{c\}, \quad a = f'(c) < 0, \quad a \neq -1, \quad P(a) = 0. \quad (125)$$

Put $\sigma = 1$ if $|a| < 1$ and $\sigma = -1$ if $|a| > 1$. Then $f^{\circ(\sigma n)}(x) \rightarrow c$ for every $x \in \mathbb{R}$ as $n \rightarrow +\infty$. Near c , the map $f^{\circ\sigma}$ is a strict Lipschitz contraction.

Proof. The inverse is C^1 and f' is negative everywhere by Lemma 26.1. Suppose the interval $[\alpha, \beta] = \text{Fix}(f^{\circ 2})$ is nondegenerate, and put $g = f^{\circ 2}$. Since $g = \text{id}$ on this interval, differentiability at β gives $g'(\beta) = 1$; the inverse derivative is also 1. Choose $x > \beta$, and let $\tau \in \{1, -1\}$ be the time direction in which $g^{\circ(\tau n)}(x) \rightarrow \beta$. The positive sequence

$$u_n = g^{\circ(\tau n)}(x) - \beta$$

is a recurrence sequence with limit zero. Lemma 26.3, applied to the even subsequence of the original orbit, says that all its active roots have modulus less than 1. Therefore $u_n = O(\rho^n)$ for some $\rho < 1$. Differentiability at β , however, gives $u_{n+1}/u_n \rightarrow 1$. A positive sequence with this ratio limit has $u_n^{1/n} \rightarrow 1$, contradicting the exponential upper bound. Hence the interval is a singleton.

If $a = -1$, the identical argument at c , where $g'(c) = a^2 = 1$, gives the same contradiction. Thus $a \neq -1$. Differentiating $P[f] = 0$ at the fixed point gives $P(a) = 0$.

The common convergent time direction already exists by Theorem 27.1. In that direction the successive ratios of the positive even distances tend to $|a|^{2\tau}$. They cannot exceed 1 in the limit, and equality was just

ruled out. Thus $|a|^\tau < 1$ and $\tau = \sigma$. Continuity of the derivative makes $f^{\circ\sigma}$ a local strict contraction. The two-cycle interval is a singleton, so both parities of every orbit converge to c in this direction. $\square$

31.2 An explicit regularity threshold

For a nonconstant P with $P(0) \neq 0$, write

$$r_-(P) = \min_{\lambda \in \text{Roots}(P)} |\lambda|, \quad r_+(P) = \max_{\lambda \in \text{Roots}(P)} |\lambda|.$$

For each negative real root $a \neq -1$, define

$$\Theta_P(a) = \begin{cases} \frac{\log r_-(P)}{\log |a|}, & -1 < a < 0, \\ \frac{\log r_+(P)}{\log |a|}, & a < -1. \end{cases} \quad (126)$$

Both quantities, when used, are at least 1. Let $\Theta(P)$ be their maximum; set $\Theta(P) = 0$ when there is no such negative root. Choose any integer

$$N \geq 1, \quad N > \Theta(P). \quad (127)$$

For a nonzero constant P , set $\Theta(P) = 0$; its solution class is empty. For example, $N = \lfloor \Theta(P) \rfloor + 1$ works. This is a sufficient threshold, not a claim that the least sufficient regularity has been found in every case. Roots that can already be removed by Theorem 30.1 may be removed before computing it, to improve the bound.

Lemma 31.2 (Finite jets detect the whole composition space). *Let f be a non-involutive C^N decreasing solution, with N as in (127). Center it at its fixed point:*

$$g(x) = f(c + x) - c, \quad V = \text{span}_{\mathbb{R}}\{\text{id}, g, g^{\circ 2}, \dots\}.$$

The map taking a function in V to its Taylor coefficients of orders $1, \dots, N$ at 0 is injective.

Proof. Evaluation at the fixed point gives $cP(1) = 0$, so $P[g] = 0$. Every function in V vanishes at 0, and right composition $E_g: v \mapsto v \circ g$ is invertible on V . Its minimal polynomial divides P . Let $a = g'(0)$ and σ be as in Proposition 31.1; set

$$b = |a|^\sigma < 1, \quad \omega = \min_{\lambda \in \text{Roots}(P)} |\lambda|^\sigma = \begin{cases} r_-(P), & \sigma = 1, \\ 1/r_+(P), & \sigma = -1. \end{cases} \quad (128)$$

Condition (127) gives $b^N < \omega$. Choose q with $b < q < 1$ and $q^N < \omega$.

Suppose $v \in V$ has zero derivatives through order N at 0. Taylor's theorem gives $v(y) = O(|y|^N)$ near zero. Every orbit $g^{\circ(\sigma n)}(x)$ eventually enters a neighborhood on which $g^{\circ\sigma}$ has Lipschitz constant at most q . Hence, for each fixed x ,

$$|g^{\circ(\sigma n)}(x)| \leq C_x q^n, \quad v(g^{\circ(\sigma n)}(x)) = O(q^{Nn}).$$

The latter is a recurrence sequence whose possible roots are λ^σ , with λ a root of P . All have modulus at least ω . By Lemma 26.3(iii), the sequence must be zero. In particular $v(x) = 0$. Since x was arbitrary, $v = 0$. $\square$

The proof is the finite-regularity replacement for an analytic identity theorem. A nonzero element of the composition space cannot vanish too rapidly at the fixed point: exponential contraction would make its orbit decay faster than any permitted recurrence mode.

Theorem 31.3 (Finite-smoothness polynomial-coordinate theorem). *Let N satisfy (127). Every C^N decreasing solution of $P[f] = 0$ is either an involution or has the unique global representation (105). In the latter case it is real analytic, real algebraic, and globally bi-Lipschitz, with exact minimal polynomial (106). In particular, Theorem 25.2 holds for C^∞ solutions.*

Proof. Suppose the solution is not an involution and retain the notation of Lemma 31.2. The order-of-vanishing filtration on V is separated by the first N derivatives. Each successive graded piece has dimension at most one: its first nonzero Taylor coefficient is a scalar. Thus V has a basis of distinct vanishing orders

$$K \subseteq \{1, \dots, N\}, \quad 1 \in K.$$

Right composition by $g(x) = ax + o(x)$ preserves this filtration and acts on the order- k graded piece by a^k . The matrix of E_g is triangular with diagonal entries a^k , $k \in K$. These entries are distinct, since $|a| \neq 1$. Therefore E_g is diagonalizable over $\mathbb{R}$. An eigenfunction with eigenvalue a^k has vanishing order exactly k , as comparison of its leading Taylor coefficient shows.

Choose the eigenfunction ψ with eigenvalue a , normalized by $\psi'(0) = 1$. Then

$$\psi(0) = 0, \quad \psi \circ g = a\psi. \quad (129)$$

Its derivative is positive near zero. Differentiating the iterated identity at an arbitrary point gives

$$\psi'(g^{\circ n}(x))(g^{\circ n})'(x) = a^n \psi'(x) \quad (n \in \mathbb{Z}). \quad (130)$$

Take $n = \sigma j$ with j large enough to enter that neighborhood. Both $(g^{\circ n})'(x)$ and a^n have sign $(-1)^n$, and neither vanishes. Therefore $\psi'(x) > 0$ everywhere. Its range is an interval. Equation (129), in the opposite time direction, produces arbitrarily large positive and negative values from any nonzero value of ψ . Its range is therefore all of $\mathbb{R}$. Thus ψ is a global C^N diffeomorphism. Let $H = \psi^{-1}$; then $H(0) = 0$, $H'(0) = 1$, and $H' > 0$.

For an eigenfunction v_k with eigenvalue a^k , set $w_k(t) = v_k(H(t))$. It obeys

$$w_k(at) = a^k w_k(t), \quad w_k(t) = \gamma_k t^k + o(t^k) \quad (t \rightarrow 0), \quad \gamma_k \neq 0.$$

For fixed t , use $a^{\sigma j} t \rightarrow 0$ to obtain

$$w_k(t) = a^{-k\sigma j} w_k(a^{\sigma j} t) \longrightarrow \gamma_k t^k.$$

The left side is independent of j , so $w_k(t) = \gamma_k t^k$ for every real t . Expand id in the eigenbasis of E_g and compose with H . The result is a polynomial identity

$$H(t) = \sum_{k \in K} h_k t^k. \quad (131)$$

Every h_k is nonzero: if the cyclic vector id missed an eigenspace, all its iterates would miss it, contradicting the definition of V . Since H is an increasing polynomial bijection, its degree is odd and its leading coefficient is positive. Equation (129) now yields (105) globally.

The fixed point c and multiplier a are intrinsic. For uniqueness of H , if H_1, H_2 are two normalized coordinates, then $L = H_1^{-1} \circ H_2$ commutes with $t \mapsto at$ and $L(t) = t + o(t)$ at zero. The same scaling argument gives $L(t) = t$ everywhere. Hence $H_1 = H_2$.

In the t coordinate, the scalar iterates are

$$f^{\circ n}(c + H(t)) = c + \sum_{k \in K} h_k a^{kn} t^k. \quad (132)$$

Distinct a^k and the independent functions t^k give the exact minimal polynomial (106); the constant mode is present precisely when $c \neq 0$. Since $H' > 0$, the inverse function theorem makes H^{-1} real analytic at every real point. Thus f is real analytic. Eliminating t from $x = c + H(t)$ and $y = c + H(at)$ shows that f is real algebraic. Finally,

$$f'(c + H(t)) = a \frac{H'(at)}{H'(t)}.$$

Its absolute value is positive and continuous, and tends to $|a|^{\deg H}$ as $t \rightarrow \pm\infty$. It therefore has positive lower and finite upper bounds, proving global bi-Lipschitz regularity. $\square$

Corollary 31.4 (Automatic analyticity with the exact exception). *If P is not divisible by $z^2 - 1$, every decreasing C^N solution, with N as in (127), is real analytic. If $z^2 - 1$ divides P , nonanalytic C^∞ involutions are genuine exceptions.*

Proof. The theorem handles non-involutions. An involution has minimal polynomial $z + 1$ or $z^2 - 1$; without the second divisor only $-\text{id}$ can occur. For the exception, take a nonzero smooth bump η supported in $(1, 2)$, small enough that $\psi(x) = x + \eta(x)$ has positive derivative. Then $f = \psi^{-1} \circ (-\text{id}) \circ \psi$ is a smooth decreasing involution. It equals $-x$ outside a compact set but is not $-x$ everywhere, so the real-analytic identity theorem shows that it is not globally real analytic. $\square$

Corollary 31.5 (Spectral deletion under sufficient regularity). *Every nonreal root and every repeated root can be deleted from a universal annihilator for decreasing C^∞ solutions, and also for decreasing C^N solutions at the threshold (127). In particular all their minimal polynomials are squarefree and real-rooted.*

Proof. For non-involutions use (106); for involutions use $\mu_f \mid z^2 - 1$. $\square$

31.3 Algebraic does not mean rational

Corollary 31.6 (Rational solutions are affine). *A decreasing rational function defined on all of $\mathbb{R}$ that satisfies $P[f] = 0$ is affine.*

Proof. Its rational iterates span a finite-dimensional space. A common denominator for a fixed finite basis shows that their degrees as rational maps are uniformly bounded. If d is the degree of f as a rational map of the complex projective line, its n th iterate has degree d^n . For completeness, degree multiplicativity follows by homogenizing coprime numerator–denominator pairs: substitution of a pair of degree e into a pair of degree d yields degree de and no common projective zero, hence no cancellation. Boundedness therefore forces $d = 1$. A real degree-one rational map with a nonconstant denominator has a real pole. Since f is defined on all of $\mathbb{R}$, its denominator must be constant. Thus f is affine. $\square$

The polynomial-coordinate theorem supplies many nonlinear algebraic solutions, but this corollary prevents them from simplifying to nonlinear rational functions. The degree of H bounds the algebraic degree of f over $\mathbb{R}(x)$; that bound need not be asserted to be sharp in every specialization.

32 Exact smooth spectra and their universal core

32.1 Which integer-power spectra actually occur?

Theorem 32.1 (Smooth spectral occurrence). *Apart from $z + 1$ and $z^2 - 1$, the polynomials occurring as decreasing C^∞ minimal spectra are exactly*

$$(z - 1)^{\varepsilon_+} \prod_{k \in K} (z - a^k), \quad \varepsilon_+ \in \{0, 1\}, \quad a < 0, \quad a \neq -1, \quad (133)$$

where K is a finite set of positive integers containing 1 and having odd maximum. Each is realized by a globally real-analytic, globally bi-Lipschitz map. Thus the C^∞ and real-analytic classes have exactly the same possible minimal polynomials.

Proof. Necessity follows from Theorem 31.3 applied to $P = \mu_f$. For sufficiency, let $M = \max K$. If $M = 1$, take $H(t) = t$. For odd $M \geq 3$, start with

$$H_0(t) = t + t^M, \quad H'_0(t) = 1 + Mt^{M-1} > 0.$$

For every $1 < k < M$ in K , the function $k|t|^{k-1}/(1 + M|t|^{M-1})$ is bounded on $\mathbb{R}$. Add sufficiently small nonzero multiples of these t^k to H_0 . The derivative stays positive, the support becomes exactly K , and the normalization $H'(0) = 1$ is unchanged. Define $f(c + H(t)) = c + H(at)$, taking $c = 0$ when $\varepsilon_+ = 0$ and $c = 1$ when $\varepsilon_+ = 1$. The exact formula (132) proves the claimed minimum. The two involution spectra have analytic reflection realizations. $\square$

The odd-maximum condition is indispensable. For example, the support $\{1, 2\}$ cannot be the support of an increasing polynomial bijection of $\mathbb{R}$, while $\{1, 2, 3\}$ can. The roots a^k are not arbitrary real roots of the correct signs: the same multiplier a must account for every one of them by an integer exponent.

32.2 The exact smooth reduction formula

Let $\varepsilon_+ = \mathbf{1}_{P(1)=0}$. For each negative root $a \neq -1$ of P , form the finite set

$$K_P(a) = \{k \geq 1 : P(a^k) = 0\}. \quad (134)$$

It is finite because the powers of a are distinct, and it contains 1. Let $M_P(a)$ be its largest odd member, and set

$$J_P(a) = K_P(a) \cap \{1, \dots, M_P(a)\}, \quad B_a(z) = (z - 1)^{\varepsilon_+} \prod_{k \in J_P(a)} (z - a^k). \quad (135)$$

If $P(-1) = 0$, also form the involution branch

$$B_{\text{inv}}(z) = (z - 1)^{\varepsilon_+} (z + 1). \quad (136)$$

Let $\mathcal{B}(P)$ consist of all the polynomials in (135) and, when present, (136).

Theorem 32.2 (Universal smooth core). *The polynomial*

$$\boxed{C_\infty(P) = \text{lcm}_{B \in \mathcal{B}(P)} B} \quad (137)$$

with the empty least common multiple interpreted as 1, satisfies

$$[P[f] = 0 \Rightarrow Q[f] = 0 \text{ for all decreasing } C^\infty \text{ maps } f] \iff C_\infty(P) \mid Q. \quad (138)$$

The same polynomial works for globally real-analytic solutions and for C^N solutions whenever $N > \Theta(P)$. Every branch $B \in \mathcal{B}(P)$ is itself attained as an exact real-analytic minimal polynomial, and every solution in any of these regularity classes has minimum dividing at least one branch.

Proof. For a non-involutive smooth solution, its multiplier a is a negative root of P . Its coordinate support K lies in $K_P(a)$, contains 1, and has odd maximum. Therefore $K \subseteq J_P(a)$, and its minimal polynomial divides B_a . Conversely, $J_P(a)$ itself has odd maximum, so Theorem 32.1 realizes the whole branch B_a . Involutions are exactly accounted for by B_{inv} . Taking the least common multiple proves the universal property and its sharpness. The finite-smoothness theorem says that the C^N class introduces no additional non-involutive spectra. Any additional nonanalytic involutions still have minimum dividing B_{inv} . Since all maximal branches have analytic realizations, the core is unchanged. $\square$

Remark 32.3 (Exact computation versus floating-point root plots). Formulas (134)–(135) use equality of real numbers, not a tolerance test on approximate logarithms. For a polynomial with rational or algebraic coefficients, root isolation and exact algebraic arithmetic can decide the finitely many tests. A finite exponent bound comes from the extreme moduli of P : for $|a| > 1$, only $k \leq \log r_+(P)/\log |a|$ can occur; for $|a| < 1$, only $k \leq \log r_-(P)/\log |a|$ can occur. Certified overestimates suffice. The formulas do not promise a particular bit-complexity for the root isolation step.

32.3 Smooth witness complexity is unbounded

Because (137) is squarefree, the least number of smooth witnesses is the minimum number of branch root sets needed to cover $\text{Roots}(C_\infty(P))$. This is an exact finite covering formulation: every solution is contained in one branch, and every branch can be realized. Branches contained in larger branches may first be discarded. No general complexity classification of this special covering problem is needed for the following sharp contrast with Corollary 30.3.

Theorem 32.4 (One continuous witness versus arbitrarily many smooth witnesses). *For every positive integer m , there is a polynomial P_m whose continuous decreasing core is attained by one globally bi-Lipschitz C^1 solution, but whose smooth core requires exactly m witnesses. Indeed, let*

$$r_j = 2 + \frac{j}{m+1} \quad (1 \leq j \leq m), \quad P_m(z) = \prod_{j=1}^m (z + r_j). \quad (139)$$

Its only decreasing C^2 solutions are $f(x) = -r_j x$, $1 \leq j \leq m$. For $m \geq 2$, a single exact-core C^1 realization is not C^2 .

Proof. All roots of P_m are negative, distinct, and outside the unit circle, so the continuous core is P_m itself. Since $2 < r_j < 3$, no positive integer power of r_j of exponent at least two belongs to the interval $(2, 3)$. Hence $K_{P_m}(-r_j) = \{1\}$, and each smooth branch is just $z + r_j$. Also

$$\Theta(P_m) = \frac{\log r_m}{\log r_1} < \frac{\log 3}{\log 2} < 2.$$

Theorem 31.3 therefore applies already at C^2 . There is no unit root, so its only C^2 solutions are the stated homogeneous linear maps. Their distinct singleton spectra show that exactly m witnesses are required.

For the C^1 realization, set $h(t) = \sum_{j=1}^m r_j^t$ and use (113). This positive increasing coordinate has exact shift spectrum $\{r_1, \dots, r_m\}$, and the resulting F has minimum P_m . Its derivative on the positive ray is

$$F'(h(t)) = -\frac{\sum_j (\log r_j) r_j^{t+1}}{\sum_j (\log r_j) r_j^t},$$

a negative weighted average of the r_j . As $t \rightarrow -\infty$, both $F(h(t))/h(t)$ and $F'(h(t))$ tend to $-r_1$. Odd extension makes F a global C^1 diffeomorphism. If $m \geq 2$, put $\gamma = \log r_2 / \log r_1 \in (1, 2)$. Writing $x = h(t)$ at the finite end gives

$$F(x) + r_1 x = -\sum_{j=2}^m (r_j - r_1) r_j^t \sim -(r_2 - r_1) x^\gamma \quad (x \downarrow 0).$$

This is incompatible with the $O(x^2)$ remainder of a C^2 function with value 0 and derivative $-r_1$ at zero. Thus the realization is not C^2 . $\square$

The witness-count difference is not due to root multiplicities, nonreal roots, opposite sides of the unit circle, or a nonzero fixed point: none of those occurs in (139). It is caused solely by the common-multiplier constraint imposed by smoothness.

33 Examples, sharp boundaries, and exact certificates

33.1 A classical benchmark: contraction or expansion, not both

Take

$$P(z) = (z + 2)(z + \tfrac{1}{2}).$$

The two continuous branch cores are $z + 2$ and $z + \frac{1}{2}$, so the only decreasing solutions are $-2x$ and $-x/2$. Their least common multiple is P , but no single solution has that minimal polynomial. This is the quadratic example behind Draga's Remark 4 [2], not a new counterexample. It illustrates why a sharp universal annihilator need not itself be a minimal polynomial of one solution.

33.2 A nondegenerate two-cycle interval

For $b > 0$ and $r > 0, r \neq 1$, define

$$F(x) = \begin{cases} -x, & |x| \leq b, \\ -rx + (r-1)b \operatorname{sgn}(x), & |x| > b. \end{cases} \quad (140)$$

It is odd, decreasing, and globally bi-Lipschitz. Its square fixes exactly $[-b, b]$. For $x > b$,

$$F^{on}(x) = (-1)^n (b + r^n(x - b)),$$

so its exact minimal polynomial is $(z+1)(z+r)$. At the endpoints, the magnitudes of the one-sided derivatives are 1 and r . Thus the map is not C^1 , in agreement with Proposition 31.1. The verifier checks the identity for $b = 1, r = 2$ at 401 rational points, while the displayed orbit formula proves it for all real inputs.

33.3 Nonreal roots really survive in the decreasing class

Let

$$h(t) = 2^t \left(1 + \frac{1}{100} \cos \frac{2\pi t}{3} \right), \quad F(h(t)) = -h(t+1), \quad F(-x) = -F(x). \quad (141)$$

The derivative of h is positive: the constant term in its bracket is $\log 2$, and the sum of the absolute perturbation amplitudes is at most $(\log 2 + 2\pi/3)/100 < \log 2$. The limits at the two ends make h an increasing bijection onto $(0, \infty)$. Its exact shift polynomial is $z^3 - 8$, with three nonzero spectral components. Consequently

$$\mu_F = z^3 + 8 = (z+2)(z^2 - 2z + 4), \quad F^{\circ 3}(x) = -8x. \quad (142)$$

Its orbit beginning at $h(0)$ has exact first values

$$\frac{101}{100}, \quad -\frac{199}{100}, \quad \frac{199}{50}, \quad -\frac{202}{25}, \quad \frac{398}{25}, \quad -\frac{796}{25}.$$

The 3×3 Hankel determinant of its first five values is $-27/625 \neq 0$, independently certifying that the orbit has no recurrence of order below three (the argument of Section 23). The map is globally bi-Lipschitz and analytic off zero, but is not differentiable at zero: the quotient $F(h(t))/h(t)$ is a nonconstant three-periodic function of t , so it has no limit as $t \rightarrow -\infty$. Consistently, $\Theta(z^3 + 8) = 1$ gives a C^2 obstruction from the finite-smoothness theorem as well.

This construction is the orientation-reversed counterpart of Part I's increasing cubic mechanism (Theorem 2.1, with minimal polynomial $z^3 - 8$). Its value here is as a transparent sharpness example for the new classification, not as the principal novelty of Part III.

33.4 Multiplicity is constrained at the extremes, not inside

For every positive integer m , the polynomial

$$R_m(z) = (z+2)(z^2+9)^m(z+4) \quad (143)$$

is negative-admissible. Its negative extreme roots have multiplicity one, while the interior nonreal roots $\pm 3i$ have arbitrary multiplicity m . Thus R_m is realized by a globally bi-Lipschitz decreasing map. No sufficiently smooth solution has this exact minimum. In fact $\Theta(R_m) = \log 4 / \log 2 = 2$, so C^3 already rules it out, regardless of m .

At two distinct extreme radii, even multiplicities are permitted:

$$(z+2)^2(z+4)^2$$

is realizable, whereas $(z+2)^2$ is not a minimal decreasing spectrum. For the former, an explicit positive envelope is

$$w(t) = (1-t)2^t + (1+t)4^t > 0.$$

On $[-1, 1]$ both summands are nonnegative. Outside that interval, the ratio of a negative summand to the positive one is less than $1/2$. Integrating from $-\infty$ gives

$$h(t) = 2^t \left(\frac{1-t}{\log 2} + \frac{1}{(\log 2)^2} \right) + 4^t \left(\frac{1+t}{\log 4} - \frac{1}{(\log 4)^2} \right),$$

and (113) realizes the claimed multiplicities exactly. The parity condition belongs to a *single* circle, not independently to each of two extreme circles. (Compare Part II's increasing example $(z-2)^2(z-4)^2$, Section 19.)

33.5 A nonlinear analytic map with a positive characteristic root

Let

$$H(t) = t + \frac{1}{4}t^2 + t^3, \quad H'(t) = 3 \left(t + \frac{1}{12} \right)^2 + \frac{47}{48} > 0. \quad (144)$$

Define $f(H(t)) = H(-2t)$. Then f is decreasing, globally analytic, and nonlinear, with

$$\mu_f = (z+2)(z-4)(z+8) = z^3 + 6z^2 - 24z - 64. \quad (145)$$

Thus smooth rigidity does not mean that every root must be negative. The positive root $4 = (-2)^2$ is an allowed even power of the common negative multiplier. Its orbit at $H(1) = 9/4$ begins

$$\frac{9}{4}, \quad -9, \quad 72, \quad -504, \quad 4176.$$

The corresponding 3×3 Hankel determinant is $46656 \neq 0$. The exact polynomial identity

$$H((-2)^3 t) + 6H((-2)^2 t) - 24H(-2t) - 64H(t) = 0$$

is checked coefficient by coefficient by the shipped verifier. A nonzero translation of the fixed point adds exactly the simple factor $z - 1$.

Using the same nonodd H with multiplier -1 gives a nonlinear analytic involution. It has minimum $z^2 - 1$, not $z + 1$: if it were $-\text{id}$, the defining identity would force H to be odd. This is why the involutive exception must be retained in the smooth theorem.

33.6 A sharp C^1 versus C^2 boundary

An especially simple non-polynomial coordinate is

$$h(t) = 2^t + 3^t, \quad F(h(t)) = -h(t+1), \quad F(-x) = -F(x).$$

Its exact minimum is $(z+2)(z+3)$. Its derivative on the positive ray lies between -3 and -2 and tends to -2 at zero, so the odd extension is C^1 . Write $y = 2^t$ and $\gamma = \log 3 / \log 2 \in (1, 2)$. Then

$$x = y + y^\gamma, \quad F(x) = -2x - y^\gamma = -2x - x^\gamma + o(x^\gamma) \quad (x \downarrow 0).$$

It is not C^2 . Conversely $\Theta((z+2)(z+3)) = \log 3 / \log 2 < 2$, so every decreasing C^2 solution is polynomial-coordinate and the smooth-core formula reduces it to $-2x$ or $-3x$. This verifies the sharpness of the sufficient threshold in this particular equation; it does not prove that the bound is optimal for all input polynomials. (Part I's Section 9 records the corresponding C^1 versus $C^{1,\alpha}$ boundary for increasing maps on one circle.)

34 Verification, effective use, and a formalization route

34.1 What the exact computations check

The verifier code/03-decreasing-core-verify.py (delivered as code/verify.py; the file of that name here is Part I's verifier) uses only Python's standard library and exact rational arithmetic. It represents a root profile by its radius, its type (negative real, other real, conjugate pair, or unit root), and its multiplicity. Two deliberately separate procedures are compared: a direct admissibility predicate and the closed universal-core construction of Section 30. For each input profile, every divisor profile is enumerated; the least common multiple of all admissible divisors must equal the proposed closed core. The attained branch and one-versus-two-witness criteria are checked as well.

Atlas	Input profiles	Divisor tests
Squarefree, two sides and unit roots	1,024	59,049
Single-circle multiplicities	1,125	121,500
Interior and boundary multiplicities	288	12,600
Two-sided multiplicities	256	10,000
Total across the four atlases	2,693	203,149

Every test passed. A separate exact-rational smooth-spectrum atlas checks 1,024 input root sets and 59,049 candidate divisors against the integer-power and odd-maximum formula of Theorem 32.2; all of those checks passed as well. The table above counts only the continuous-core checks. These counts are test instances across the four atlases, not a claim that all profiles across different atlases are distinct. The script also checks the two cubic Hankel determinants, the polynomial-coordinate identity, the rational plateau sample, and an eight-slope instance of the witness family (139). Recorded output is supplied in both `data/03-decreasing-core-verification.json` and `data/03-decreasing-core-verification.txt` (delivered as `results/verification.json` and `results/verification.txt`; the two files are byte-identical, and both hold the JSON record). (Merge note: in the write phase, 29 September 2026, Python 3.14.4, the verifier was rerun in a scratch copy with its delivered name restored, in about 2 seconds; it passed, and its JSON output and its printed output equal the shipped records apart from Windows line endings. It also passes under `python -O`, as its checks raise explicit exceptions rather than use `assert`.)

These are finite consistency checks. In particular, the profile predicate encodes the theorem’s stated conditions; exhaustive agreement with it does not establish their equivalence to existence of a real map. The all-real construction, the spectral necessity, and the finite-smoothness theorem rest on the proofs in Part III, not on numerical experimentation or exhaustive sampling.

34.2 A practical decision workflow

For a polynomial with certified algebraic roots, the following workflow produces exact answers. First, test for a negative real root; without one, the decreasing solution class is empty. Next construct $R_<$, $R_>$, U using exact modulus comparisons and multiplicities. This gives the continuous core, its maximum one or two branches, and the affine-rigidity test. To restrict regularity, compute a certified upper bound for $\Theta(P)$ and choose an integer strictly above it. The finite-smoothness theorem then reduces the spectral problem to the power tests $P(a^k) = 0$. These tests generate the branches B_a and the involution branch, and their least common multiple gives the smooth core.

The construction of a bi-Lipschitz witness is effective in principle: choose the finite exponential modes, enlarge Λ until the envelope’s two tail inequalities hold, and choose a positive perturbation smaller than a certified domination bound. The proofs establish termination of these searches, but the supplied checker does not implement a general certified transcendental inequality engine. An implementation of that engine would be a separate deliverable, not something hidden behind the exact finite test counts.

34.3 A possible Lean development

A formal proof should begin with the right-composition module, not with an attempt to normalize nested nonlinear expressions. The natural algebraic layer consists of a finite-dimensional cyclic space, an invertible composition operator, orbit evaluation maps, and the least-common-multiple description of the minimal

polynomial. The recurrence layer can then establish the exponential-polynomial basis and its growth lemmas. The order-topological layer proves the two-cycle interval and the common finite-end direction.

The continuous occurrence theorem additionally needs the positive envelope, domination of finitely many exponential-polynomial modes, and the analytic inverse function theorem on rays. The finite-smoothness result takes a different route after the recurrence layer: local contraction, Taylor remainder bounds, finite-jet injectivity, diagonalization by distinct vanishing orders, and global propagation of an eigenfunction coordinate.

A useful formalization milestone would be the finite-smoothness polynomial-coordinate theorem conditional only on an abstract recurrence growth lemma. A second milestone would discharge that lemma. A third would formalize the root-profile algorithms and show that their returned polynomials have the universal properties. This separation makes the trusted assumptions visible. No Lean source is included in the delivered package, and no existing repository formalization is claimed to certify these results. (As of 29 September 2026 none of these milestones, nor any stage of Part II's plan, has been started in ProveIt.)

35 Further research questions

The following problems are proposed continuations, not assertions that no related special case has ever been studied. They distinguish questions left open by Part III from statements already proved here.

35.1 Optimal finite differentiability

Determine the least integer $N_{\text{opt}}(P)$ such that every decreasing $C^{N_{\text{opt}}(P)}$ solution is either an involution or polynomial-coordinate. The bound $N > \Theta(P)$ is explicit but can overcount roots that are incompatible with a particular multiplier. A stronger algorithm should combine the continuous branch reduction, the derivative multiplier, and exact integer-power resonances before assigning a threshold. The equation $f(x) + 2x = 0$ is already rigid at C^0 , whereas the two-root examples in Section 33 show that C^1 genuinely fails where C^2 succeeds.

35.2 Complete intermediate-regularity spectra

Classify minimal spectra for each fixed C^k class below the automatic analyticity threshold, and for Hölder or Zygmund regularity. The coordinate terms suggest a mixture of fractional powers, logarithmic powers from repeated roots, and oscillations in $\log |x - c|$ from nonreal roots. A complete theorem must handle cancellations at the same asymptotic degree and the matching of the two sides of the fixed point; an inspection of one mode is not enough.

35.3 All continuous solutions, rather than only their spectra

The continuous theorem classifies which minimal polynomials occur and constructs a witness, but it does not parametrize every continuous solution with a given minimum. Develop a complete parameter space for such solutions, including the possible involution on the central interval and compatible data on an exterior fundamental interval. The recurrence equation makes that data highly constrained, while monotonicity imposes an infinite family of inequalities.

35.4 Optimal distortion and quantitative realizations

For an admissible μ , minimize a natural bi-Lipschitz distortion such as $\text{Lip}(f) \text{Lip}(f^{-1})$ over its realizations. Determine which spectral features force a positive gap from the best affine distortion, and whether the infimum is attained. The existing construction gives finite bounds but is not optimized; small spectral coefficients can preserve exact minimality while making a mode difficult to detect numerically.

35.5 Certified construction from algebraic input

Turn the existence construction into a certificate-producing program. Input should be a polynomial with algebraic coefficients and isolated roots; output should include explicit envelope parameters, exact certified domination bounds, a coordinate expression, and a proof object for the Lipschitz constants. The symbolic spectral test is finite and elementary after root data are supplied, but certification of the global exponential inequalities is a separate analytic task.

35.6 The special covering problem for smooth witnesses

Determine the complexity of the minimum branch-cover problem from Section 32. The root sets are not arbitrary: they arise from integer powers of negative anchors and an odd-maximum cutoff. This arithmetic structure might allow a simpler algorithm than general set cover. Theorem 32.4 proves that no uniform bound on the answer exists, but says nothing about the complexity of computing that answer.

35.7 Algebraic degree and exceptional polynomial coordinates

For $f = H \circ (a \text{id}) \circ H^{-1}$, determine the exact degree of f over $\mathbb{R}(x)$ from H and a . The upper bound $\deg H$ can have special reductions when the coordinate has compositional structure. Classify these reductions, and characterize when two polynomial-coordinate descriptions yield algebraically related maps. The normalized coordinate of a fixed non-involutive map is already unique; the unresolved issue concerns algebraic degree and families, not that uniqueness theorem.

35.8 Higher-dimensional ordered systems

Investigate analogues for maps preserving or reversing a proper cone in $\mathbb{R}^d$ and satisfying a finite linear relation among iterates. The one-dimensional proof relies on a finite two-cycle limit and a one-dimensional graded jet at each order. Both features change in higher dimensions: bounded recurrent motion need not collapse, and resonant jets have multidimensional eigenspaces. Identify hypotheses under which a finite-dimensional composition space still forces an algebraic global conjugacy.

35.9 Nonhomogeneous equations and stability

Extend the classification where possible to $\sum_{j=1}^d a_j f^{\circ j}(x) = F(x)$ with prescribed non-linear F , and to approximate equations on invariant domains. Existing existence and stability programs, including [7, 8], operate in this broader setting. An appropriate first target is a forcing term lying in a finite-dimensional space preserved by right composition, so that a replacement for the cyclic recurrence space survives. Small residual in an equation should not be confused with small error in exact minimal spectral support.

35.10 Independent audit and formal verification

The highest-priority mathematical audit is the passage from a family of alternating orbits to a single global spectral branch, followed by the finite-jet injectivity and global-coordinate argument. The highest-priority literature audit is a full-text comparison with the older characteristic-solution and differentiability papers, not just their abstracts. The highest-priority formal task is a verified recurrence growth lemma feeding both classification theorems. These three activities address correctness, priority, and machine-checking respectively; none substitutes for the others.

36 Conclusion of Part III

Orientation reversal imposes a structure absent from the unrestricted increasing problem: a finite two-cycle interval, a common convergent time direction, and a prohibition on mixing contracting and expanding nonunit spectra. Combined with the positive bilateral recurrence criterion, this yields a complete occurrence theorem and a closed universal reduction with at most two sharp branches.

Sufficient differentiability changes the answer more radically. The two-cycle interval disappears, the fixed point is hyperbolic, and a finite jet detects the entire composition space. Its eigenfunctions then become integer powers of one global coordinate, forcing that coordinate's inverse to be a polynomial. This proves automatic analyticity and algebraicity for non-involutions at an explicit finite regularity threshold. The smooth core is therefore governed by exact integer-power compatibility, rather than merely by extreme-circle multiplicity conditions. The unbounded witness-count example separates these two mechanisms even for simple negative real roots on one side of the unit circle.

37 Source, dependency, and claim audit of Part III

Repository and companion provenance. The repository anchor was inspected at commit

afb2d1227d8bc5df3beffc1463e958db3544960b.

The relevant report is under

SetTheory/Cardinals/docs/reports/log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/

(the manuscript cites it as the GitHub tree of that path at that commit, <https://github.com/VladimirReshetnikov/ProveIt/tree/afb2d1227d8bc5df3beffc1463e958db3544960b/SetTheory/Cardinals/docs/reports/log-concavity-and-unimodality/nonreal-roots-in-iterative-equations>). Its README, status information, and indexed source excerpts identify a circular increasing classification and explicitly limited scope; at that commit it held Part I only. The separate increasing companion, printed here as Part II, was retrieved from the user's prior-file library as the LaTeX source `article(20260929-042518).tex`, including the positive-recurrence construction and its scope statement. The manuscript does not represent it as an independently refereed publication or as a file verified to be part of the pinned repository commit.

Extent of the external literature check. The relevant full arXiv text of Draga–Morawiec [1], including Problem 6.1 on preprint page 13, and Draga’s follow-up [2], including Theorem 3 and Remarks 4–6, were inspected. The former asks about nonreal maximal-modulus roots tied with a real root. The latter already proves a particular negative-root disjunction. Neither the recurrence method nor the existence of that disjunction is a priority claim of Part III.

For Yang–Zhang [6] and Li–Zhang [7], the publisher’s abstract and bibliographic record were inspected, not the subscription full text. The former describes multiple characteristic roots and linear-difference constructions; the latter concerns existence and uniqueness of decreasing solutions in a leading-coefficient problem. A recent paper of Xia–Huo–Wang–Xia [8], published in April 2026, was also checked at abstract level. It treats sufficient conditions for existence, uniqueness, and stability in a nonhomogeneous half-line setting, not a stated complete homogeneous spectral classification in its abstract.

The broader search surfaced older differentiability and linear-dependence papers whose full contents were not obtained. Therefore Part III claims a self-contained proved extension of the inspected ProveIt program, not an exhaustive determination of first publication. The source list is a record of relevant checked context, not a complete bibliography of iterative functional equations.

Logical dependencies and limitations. The main continuous theorem depends on elementary recurrence decomposition, finite trigonometric averages, one-dimensional order dynamics, exponential domination, and the inverse function theorem. The finite-smoothness theorem additionally uses Taylor remainder bounds and elementary finite-dimensional diagonalization. The universal-core formulas are consequences of these occurrence theorems. The exact computations are supplements and are not logical premises of any all-real theorem.

The domain is the whole real line and $P(0) \neq 0$. Part III does not classify arbitrary maps on arbitrary intervals, equations with a zero constant coefficient, nonhomogeneous equations, higher-dimensional systems, or every individual continuous solution. It does classify every individual non-involutive solution in the stated sufficiently smooth class. The suggested future questions are not claimed solved. All new proofs remain open to independent mathematical review. The manuscript’s own status note is shipped verbatim as 03-decreasing-core-STATUS.md.

A Source and status audit

The literature search was performed on 20 September 2026. The primary sources below were used to identify the question and delimit the claims, not to replace the proofs in this article.

The target. Draga–Morawiec [1], Problem 6.1, is present in the retrieved HTML and in the rendered page 13 of arXiv:1503.00570v2. The preprint has 14 pages. Its published bibliographic record is *Aequationes Mathematicae* 90 (2016), 935–950, with DOI 10.1007/s00010-016-0420-4. The source is about preserving the solution equation after reducing the characteristic polynomial, rather than merely preserving the existence of at least one solution.

Subsequent context. Draga’s note [2] gives another order-reduction theorem and mentions the unrestricted nonreal-root issue in its final remarks. The later paper *Means of iterates* [4] treats a particular family of iterative mean equations and uses characteristic-root analysis. A 2021 paper [5] studies a class of fourth-order

equations under restrictions on the characteristic roots. These sources show continuing work on the subject; none of the retrieved material established priority for the counterexample developed here.

Initial lead. Petrov’s question [3] was the initial candidate. Its displayed answer count was zero in the retrieved page. An unanswered question page is not, by itself, proof that no published solution exists elsewhere.

Search limitation. The search included the authors’ names, polynomial-like iterative equations, nonreal-root elimination, equal-modulus roots, and the numbered problem. Some returned results were irrelevant and were not used. The search was not a complete citation-database survey. Thus the report does not attach a first-discovery claim to the elementary construction or to the classification theorem. The complete source identifiers and this status distinction are also recorded in `notes/literature_audit.md`.

B Reproduction, selection, and exclusion record

From the archive’s top-level directory, the following commands reproduce the exact certificates and, with the optional plotting dependency, the numerical illustrations:

```
python code/verify.py
python code/analytic_demo.py --plots
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied figure files allow the PDF to be compiled without running the plotting script. No bibliography processor is required.

The exact verifier normalizes positive rational numbers by multiplication or division by eight, not by taking an approximate logarithm. This avoids wrong branch choices at powers of eight. Its independent conjugacy implementation also uses rational operations. The recurrence polynomials in the JSON certificate are stored with coefficients in *ascending* order, so, for example, $z^3 - 8$ is represented by `[-8, 0, 0, 1]`.

The random area draw was:

Number of listed areas	80
Generator call	<code>secrets.randbelow(80)</code>
Returned zero-based index	17
Selected one-based index	18
Selected area	Functional equations
Redraws	0

An entropy-backed draw is recorded rather than seed-replayed; running a new draw is not expected to reproduce the same index. The full 80-element list is in `data/selection.json`.

The excluded input was the 1,095-line file `manifest(1).tex`, with SHA-256 fingerprint

729fb6e3aef3f30c7c5284c4296681fba72b6d6af68211f178ad0bc3c5d15b41.

The file was reviewed as an exclusion list; this report does not certify the correctness or priority of its individual mathematical claims. A suggested new manifest entry is supplied separately so that future selections can exclude the present target as well.

References

- [1] S. Draga and J. Morawiec, *Reducing the polynomial-like iterative equations order and a generalized Zoltán Boros' problem*, *Aequationes Mathematicae* **90** (2016), 935–950. doi:10.1007/s00010-016-0420-4. Preprint: arXiv:1503.00570v2, 1 December 2015. Problem 6.1 is on preprint page 13.
- [2] S. Draga, *A note on the order of polynomial-like iterative equations*, *Commentationes Mathematicae* **56** (2016), 243–249. Preprint title: *A note on the polynomial-like iterative equations order*, arXiv:1604.01287v2, 14 November 2016.
- [3] F. Petrov, *Continuous linear recurrent relations*, MathOverflow question 232493, 1 March 2016. <https://mathoverflow.net/questions/232493/continuous-linear-recurrent-relations>. Retrieved 20 September 2026. [Part II, added 29 September 2026: the question's comments by I. Bogdanov, checked for Part II on 28 September 2026, contain the piecewise-linear mechanism used in Theorem 17.1.]
- [4] S. Draga and J. Morawiec, *Means of iterates*, arXiv:1801.05006v2, 19 February 2018.
- [5] S. Thangroongvongthana, V. Laohakosol, and S. Mavecha, *A class of continuous solutions of a fourth order polynomial-like iterative equation*, *Current Applied Science and Technology* **21**(3) (2021), 495–523. <https://li01.tci-thaijo.org/index.php/cast/article/view/248735>.
- [6] D. Yang and W. Zhang, *Characteristic solutions of polynomial-like iterative equations*, *Aequationes Mathematicae* **67** (2004), 80–105. doi:10.1007/s00010-003-2708-4. [Part III, added 29 September 2026.]
- [7] L. Li and W. Zhang, *Continuously decreasing solutions for polynomial-like iterative equations*, *Science China Mathematics* **56** (2013), 1051–1058. doi:10.1007/s11425-013-4589-x. [Part III, added 29 September 2026.]
- [8] C. Xia, R. Huo, X. Wang, and Z. Xia, *Asymptotically periodic and continuous solutions of a polynomial-like iterative equation*, *Aequationes Mathematicae* **100** (2026), article 44. doi:10.1007/s00010-026-01289-3. [Part III, added 29 September 2026.]